

More than 0.7947 of the zeros of the Riemann zeta function are simple and on the critical line*

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Abstract

Let $N(T, 2T)$ count the nontrivial zeros of $\zeta(s)$ with ordinates in $(T, 2T]$, and N_0^s , N_d those simple and on the critical line, respectively distinct. Subject to the verification grading stated below and in §11, we establish, with ineffective constants,

$$\liminf_{T \rightarrow \infty} \frac{N_0^s}{N} \geq 0.7947, \quad \liminf_{T \rightarrow \infty} \frac{N_d}{N} \geq 0.8973,$$

and the same for any fixed primitive Dirichlet L -function; en route, the rungs 0.6916/0.8458 (one-sided fourth-moment remainder, via Matomäki–Radziwiłł–Tao), and $13/18 = 0.7222$, $31/36 = 0.8611$ (exact fourth moment). The proof evaluates the third through sixth trace moments of the Gabor-compressed Weil matrix of [4] — $m_3 = 2$ and $m_4 = \frac{13}{4}$ exactly, $m_5 = \frac{67}{12} + C_5$ and $m_6 = 10.1657(18)$, the remainders being connected sine-kernel Ursell constants — and consumes them through a moment-pinned linear program built on an explicit Chebyshev–Markov certificate — an *exact rational* dual certificate, valid on unbounded support. The fourth-moment engine (“Lemma D”) is proved in the truncated form the program consumes: the cell ledger is ℓ_1 -truncated at $(\log X)^B$ with power-saving tails, the lock variance transports by exact product and Parseval identities to a spectral frame in which every factor is a windowed single prime sum at a rational of logarithmic denominator, and the Siegel–Walfisz theorem plus Vaughan’s estimate close all bins — no dispersion method and no large-modulus technology anywhere. The local main terms at every cycle length are pinned by a four-line coincidence-counting closed form; the Bell(5) and Bell(6) ledgers collapse onto four-cycle anchors, now certified as exact rationals ($t_{\text{adj}} = \frac{7}{60}$, $t_{\text{opp}} = \frac{1}{30}$, $\{2, 2, 2\} = \frac{131}{420}$: exact polyhedral integration, Appendix C), through machine-exact frozen-slot identities. *Claim grading*: the identity, certificate and pairing-constant layers are machine-verified or

*Verification status (read this first): the chain of this paper consists of classical inputs, machine-verified identities, certified constants, and numerically evaluated connected constants carrying stated two-sided bands. Its claim grade is *certified-candidate*, not record: the certified closed forms of C_5 , $\{4, 2\}$ and $\{6\}$ remain open — though each is proved RATIONAL (a finite signed sum of rational polytope volumes, §8.6), giving the designated exact-integration route an explicit representation — the Lean formalization covers the identity and certificate layers only (kernel-compiled, core Lean 4, zero **sorry**; §11(f)), the analytic layer is unformalized, and no external review has taken place. The three write-out items W1–W3 are written out (§8.4), the pairing-layer constants are certified as exact rationals (§8.3, Appendix C), and the consumption step is backed by an exact rational dual certificate valid on unbounded support (Lemma 9.1). The precise grading of every layer, the discrepancy register, and the program’s verification gate (formalization and external review precede any record claim) are in §11 and Appendix B.

exact-rational; the write-out items W1–W3 are written out in §8.4; the analytic chain is certified-candidate; the three connected constants $C_5 = 0.0278(1)$, $\{4, 2\} = -0.0552(8)$ and $\{6\} = -0.0078(8)$ are midpoint-grid ladder evaluations with stated bands (§8.6), proved rational by the polytope dimension theorem of §8.6 but not yet evaluated in closed form; the consumption layer (inertia, tail, counting) is that of [4], recalled in §2. The consumption step itself is exact: a rational sum-of-squares dual certificate, kernel-checked in core Lean 4 (twenty theorems across the identity, local-law and certificate layers, zero `sorry`). A standalone reproduction and certification package accompanies the paper; a falsifier registry (194 pre-registered checks, 34 fired and converted, all forward-recorded) documents the research trail.

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1 Introduction

That a positive proportion of the nontrivial zeros of $\zeta(s)$ lie on the critical line is due to Selberg [15]; Levinson [10] obtained the proportion $\frac{1}{3}$, Conrey [5] reached $\frac{2}{5}$, and the record along Levinson’s method stands at $\kappa > 5/12$ [14] (see also [3, 7]). Very recently the two-thirds theorem [4] proceeded by a different route: a linear-algebraic reading of Montgomery’s pair-correlation sum, in which spectral information is read from trace moments of a Gabor compression $\tilde{G}$ of Weil’s Hermitian form, and the first two moments alone — both computable unconditionally from Montgomery’s prime-side evaluation at bandwidth $\lambda \leq 1$ [12, 1] — already yield the proportion $2/3 - o(1)$ (and 0.6725 with the Montgomery–Taylor window [13]). The present paper extends that framework to the third through sixth moments. By [4, Rem. 1.1] no certificate consuming only the bandwidth-one data can exceed 0.68185; the route past the ceiling identified there (§7.5(f) of [4]) is the fourth moment, $\text{tr } \tilde{G}^4 = d\ell_1^4(13/4 + o(1))$, worth 13/18.

This paper climbs that ladder in four stages, in a single text with one notation, one bibliography, one verification ledger, and one claim grading: the certificate calculus and the third moment (§3), a logically independent one-sided fourth-moment route reaching 0.6916 (§4), the exact fourth moment worth 13/18 (§§5–7), and the fifth and sixth moments with their consumption (§§8–9), reaching the theorem below.

Theorem 1.1 (Main theorem; grade: certified-candidate, see §11). *With ineffective constants,*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.7947, \quad \liminf_{T \rightarrow \infty} \frac{N_d(T, 2T)}{N(T, 2T)} \geq 0.8973,$$

and the same holds for the zeros of any fixed primitive Dirichlet L-function. En route, each rung consuming exactly what its chain establishes: $m_4 \rightarrow 13/4$ two-sidedly (hence $13/18 - \varepsilon$ and $31/36 - \varepsilon$), and, by the logically independent one-sided route of §4, 0.6916/0.8458. (An intermediate fourth-and-fifth-moment rung considered at an earlier stage of this research is repaired and demoted in §9: without sixth-moment information it is not supported on unbounded spectral support — register item D6.)

The dependency structure of Theorem 1.1 is graded explicitly in §11: the 13/18 rung rests on the classical-technology proof of Lemma D (§7) and the consumption layer of [4] (recalled in §2); the 0.7947 headline additionally consumes the three connected Ursell constants of §8 as numerical evaluations with stated two-sided bands (the pairing layer is exact, §8.3). Nothing in this paper bears on the Riemann Hypothesis itself: proportion statements are insensitive to $o(N)$ off-line zeros: the inputs (functional equation, explicit formula, mean values of Dirichlet

polynomials of length $\leq T$) are satisfied by objects for which the analogue of RH is false, and the argument produces lower bounds only [4, §1.5].

Relation to [4], and what is new

This paper is built on [4] and consumes its framework without repetition: the compressed Weil matrix, the zero-side block structure, the tail estimate, and the first two trace moments are used exactly as established there (§2). The advance is fourfold. (i) *The moment ladder*: [4] consumes only $\mu_1 \rightarrow 1$, $\mu_2 \rightarrow \frac{4}{3}$; this paper evaluates $\mu_3 \rightarrow 2$ and $\mu_4 \rightarrow \frac{13}{4}$ exactly, and μ_5, μ_6 up to three connected constants with proved rationality (§§3–8). (ii) *The truncation technology*: the ℓ_1 -truncated cell ledger with Siegel–Walfisz/Vaughan closure (§5), absent from [4], is what makes the higher moments computable with classical tools. (iii) *The consumption*: the rank–trace inequality of [4, Lem. 3.2] — the matrix form of $m^2 \geq 2m - 1$ — is replaced by an exact Chebyshev–Markov dual certificate at degree six (§9), the analogue of $m^2 \geq 3m - 2$ and beyond, valid on unbounded support with exact rational corners. (iv) *The verification apparatus*: pre-registered falsifier checks, a forward-only discrepancy register, and a kernel-compiled Lean layer (§11), replacing the traditional single-grade presentation of [4]. The constants rise from $2/3$ and $\frac{5}{6}$ (distinct) to 0.7947 and 0.8973.

Relation to the pair-correlation route

Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh [1] showed that Montgomery’s pair-correlation method yields two-thirds-type conclusions under hypotheses weaker than RH, and Goldston–Suriajaya [2] reach the two-thirds constant itself conditionally; those are conditional statements about the *zeros*’ pair correlation. The present route is disjoint, and unconditional: it consumes the average behaviour of $2k$ -point correlations of $\Lambda \star \Lambda$ — prime-side objects, controlled by the Siegel–Walfisz theorem and Vaughan’s estimate alone — through the compressed-matrix framework, with no hypothesis on the zeros. The two-thirds threshold, previously reached only under such hypotheses, is crossed here unconditionally with margin ($0.7947 > 2/3$), and the additional information consumed relative to the bandwidth-one data is exactly the third-through-sixth moment chain of §§3–8.

The method in one page

The moments m_k of the spectral measure of $\tilde{G}$ are carried by k -cycle ledgers: cells indexed by prime-power modulus tuples and lock (offset) vectors, weighted $1/(d\ell_1^k)$, with window-overlap volumes supplying a walk geometry. The consumption is an explicit Chebyshev–Markov certificate (§3): mass near the origin of the spectral measure is bounded by moment data alone, with exact exchange rates. Three observations reduce the moment evaluations to classical tools.

Truncation (§5). The $1/\ell_1^k$ weights discharge all cells with $\ell_1 > (\log X)^B$ trivially and two-sidedly; only logarithmic moduli remain.

The spectral frame (§7, §8). The lock variance of a k -cycle obeys exact identities — at $k = 4$ a discrete Parseval identity; at every k the *product identity* $\sum_j N_k(j)^2 = \sum_t A_1(t) \cdots A_k(t)$ (two chains share a lock vector iff they differ by a common shift) — which exhibit it as a spectral L^2 -distance whose every factor is a *windowed single prime sum*. At logarithmic moduli every main-term frequency is a rational of logarithmic denominator: the Siegel–Walfisz theorem gives an arbitrary log-power saving pointwise on the doubly-major bins, Vaughan’s estimate closes the minor and mixed bins, and the singular-series comb is identified exactly by the local law

below. The apparent “twin hardness” of locked correlations dissolves: no per-lock evaluation is ever attempted.

The local law and the anchor collapse (§6, §8). The mod- p structure of every cycle length is an affine lock chain, whose density obeys a coincidence-counting closed form proved in four lines and verified exhaustively. The $\text{Bell}(5) = 52$ and $\text{Bell}(6) = 203$ class ledgers of the fifth and sixth moments then collapse onto the certified four-cycle anchors through machine-exact frozen-slot identities; the only genuinely new constants above m_4 are the connected Ursell values.

Verification discipline and contributions

The research program behind this paper operates falsifier-first: every structural claim carries a pre-registered numerical check; corrections are recorded forward, never edited away; claims are graded. Thirty-four of the registry’s checks have fired and been converted — several of them shaping this paper directly (Appendix B). The chain’s grade, its named write-out items, and the results of the certification audit are stated in §11; per the program’s verification gate, Lean formalization and external review precede any record claim. The mathematical development of this paper was carried out by Claude (Anthropic), a large language model, within a research program directed by the authors, who specified objectives, verification discipline, and publication decisions, and who take full academic responsibility for the content. In accordance with prevailing journal authorship policies, the model is not listed as an author; its role is stated here and in the Acknowledgements. The theorem claims of this paper are advanced *at the stated grade*: as certified candidates, not as established records.

2 Framework and notation, following [4]

The construction of the compressed Weil matrix $\tilde{G}$ — Weil’s Hermitian form, the test family, the zero-side block structure, the tail estimate, and the first two trace moments — is exactly that of the two-thirds paper [4] (§§2–5 there). This section recalls the notation and states the five results of [4] that the moment ladder consumes; proofs are in [4], whose own analytic imports are the published literature [17, 9, 16, 12, 1]. Our contribution begins with the third moment (§3).

2.1 Weil’s form, the test family, and $\tilde{G}$

Throughout, $l = \log(T/2\pi)$, $\ell_1 = l + 2\log 2 - 1$ (so $N(T, 2T) = T\ell_1/2\pi + O(l)$), and for fixed $\lambda \in (0, 1]$: $L = \lambda l$, $X = e^L$, $d = \lfloor LT/2\pi \rfloor$. For $f \in C_c^2(\mathbb{R})$ put $h_f(z) = \hat{f}(z) = \int f(u)e^{izu} du$; two integrations by parts give

$$|h_f(x + iy)| \leq e^{|y|\Lambda_f} \min(\|f\|_1, \|f''\|_1 |x + iy|^{-2}), \quad \text{supp } f \subset [-\Lambda_f, \Lambda_f]. \quad (2.1)$$

Weil’s Hermitian form is $W(f, g) = \sum_{\rho} m_{\rho} h_f(\gamma_{\rho}) \overline{h_g(\gamma_{\rho})}$, the sum over distinct nontrivial zeros $\rho = \beta + i\gamma_{\rho}$ (multiplicity m_{ρ} explicit; absolutely convergent by (2.1) and $\sum m_{\rho} |\gamma_{\rho}|^{-2} < \infty$). The zero multiset is invariant under $\rho \mapsto 1 - \bar{\rho}$, so W is Hermitian. With $s = \frac{1}{2} + i\tau$ define the prime-side densities

$$\mu(\tau) = \frac{1}{2\pi} \text{Re} \frac{\Gamma'}{\Gamma} \left(\frac{1}{4} + \frac{i\tau}{2} \right) - \frac{\log \pi}{2\pi}, \quad \Pi_X(\tau) = \frac{1}{2\pi(\frac{1}{4} + \tau^2)} + \frac{1}{\pi} \text{Re} \frac{X^s - 1}{s}, \quad P_X(\tau) = -\frac{1}{\pi} \sum_{n \leq X} \frac{\Lambda(n)}{\sqrt{n}} \cos(\tau \log n),$$

and $\nu_X = \mu + \Pi_X + P_X$. Weil's explicit formula [17] (in the normalization of [9, Thm. 5.12], applied to $k = f * \tilde{g}$, $\tilde{g}(u) = g(-u)$, whose transform $h_k = h_f h_{\tilde{g}}$ is entire with the decay (2.1)) gives the spectral form

$$W(f, g) = \int_{\mathbb{R}} h_f(\tau) \overline{h_g(\tau)} \nu_X(\tau) d\tau, \quad \text{supp } f, g \subset [-\frac{L}{2}, \frac{L}{2}]. \quad (2.2)$$

We record $|\Pi_X(\tau)| \leq 3\sqrt{X}/(1 + |\tau|)$, $|P_X(\tau)| \ll \sqrt{X}$, $\int_T^{2T} \mu = N(T, 2T) + O(l)$ and $\int_T^{2T} \mu^2 = T\ell_1^2/4\pi^2 + O(l^2)$ (Stirling).

Fix a nondecreasing $\varrho \in C^3(\mathbb{R})$ with $\varrho = 0$ on $(-\infty, 0]$, $\varrho = 1$ on $[1, \infty)$, a ramp width $1 \leq w \leq L/8$, and the taper $\phi(u) = \varrho((L/2 - |u|)/w)$: even, $0 \leq \phi \leq 1$, $\text{supp } \phi = [-L/2, L/2]$, $\phi = 1$ on $[-L/2 + w, L/2 - w]$. Put $a = \frac{1}{L} \int \phi^2$, $b = \frac{1}{L} \int \phi^4$ (so $1 - 2w/L \leq b \leq a \leq 1$), $\Phi = \widehat{\phi^2}$, and note the window calculus

$$(L - 2w - |y|)_+ \leq g(y) := (\phi^2 \star \phi^2)(y) \leq (L - |y|)_+. \quad (2.3)$$

The Gabor system at critical density is $f_k(u) = \phi(u)e^{-i\tau_k u}$, $\tau_k = T + 2\pi k/L$, $0 \leq k < d$, and the compressed Weil matrix is

$$G_{kl} = W(f_k, f_l) = \sum_{\rho} m_{\rho} \widehat{\phi}(\gamma_{\rho} - \tau_k) \widehat{\phi}(\gamma_{\rho} - \tau_l) = \int_{\mathbb{R}} \widehat{\phi}(\tau - \tau_k) \widehat{\phi}(\tau - \tau_l) \nu_X(\tau) d\tau, \quad \tilde{G} = G/L,$$

real symmetric by either expression; its normalized trace moments are $\mu_k = \text{tr } \tilde{G}^k / (d\ell_1^k)$.

Lemma 2.1 (Poisson summation for the Gabor system). *For all $\tau, \tau' \in \mathbb{R}$, $\sum_{k \in \mathbb{Z}} \widehat{\phi}(\tau - \tau_k) \widehat{\phi}(\tau' - \tau_k) = L \Phi(\tau - \tau')$; in particular $\sum_{k \in \mathbb{Z}} \widehat{\phi}(\tau - \tau_k)^2 = aL^2$.*

Proof. This is [4, Lem. 2.2]: the Gabor system at critical density $h = 2\pi/L$ has a translation-invariant frame kernel with no aliasing error, for any window supported in an interval of length L . $\square$

2.2 The zero side: inertia, tail, and the counting inequality

Fix $D_0 = T^{1/2}$, $I = (T, 2T]$, $I' = (T - D_0, 2T + D_0]$, and split $G = A + E$ over distinct zeros with $\gamma \in I'$ (matrix A) and the rest (E); both are real symmetric. Classify the distinct zeros with $\gamma \in I'$: S_1 (simple, on-line; s_1), S_2 (multiple, on-line; s_2), and off-line pairs $\{\rho, 1 - \bar{\rho}\}$ (p of them), so $\#Z(I') = s_1 + s_2 + 2p$ and $N(I') \geq s_1 + 2s_2 + 2p$.

Proposition 2.2 (Block structure). *$n_+(\tilde{A}) \leq s_1 + s_2 + p$ and $\text{rank } \tilde{A} \leq \#Z(I')$.*

Proof. This is [4, Prop. 4.1]: Sylvester inertia under pull-back — an on-line zero contributes a positive 1×1 block, an off-line pair a hyperbolic block of signature $(1, 1)$, and no independence of the evaluation functionals is needed. $\square$

Proposition 2.3 (Tail). *Let A_0 be an absolute constant with $N(t + 1) - N(t) \leq A_0 \log(t + 3)$ [16, Thm. 9.2]. Then for $T \geq T_0$,*

$$\|\tilde{E}\|_{\text{op}} \leq \theta_0 := 4A_0 C_1^2 X^{1/2} \frac{\log(4T)}{D_0^2} \ll l T^{\lambda/2-1}, \quad C_1 = \|\phi''\|_1 = \frac{2\|\varrho''\|_1}{w},$$

and the trace norm satisfies $\|\tilde{E}\|_1 \leq 2\theta_0/L$.

Proof. This is [4, Prop. 4.2] (the taper's r^{-2} -decay (2.1) against the zero-density bound); we restate the explicit θ_0 because the truncation discharge of §5 and the third-moment tail step consume it. $\square$

Proposition 2.4 (Counting inequality). *For every $\theta \geq \theta_0$,*

$$s_1 \geq 2n_+^\theta(\tilde{G}) - N(I'), \quad \#Z(I') \geq n_+^\theta(\tilde{G}),$$

hence, with $N(I' \setminus I) \ll D_0 l = o(N)$,

$$N_0^s(T, 2T) \geq 2n_+^\theta(\tilde{G}) - N(T, 2T) - 2N(I' \setminus I), \quad N_d(T, 2T) \geq n_+^\theta(\tilde{G}) - N(I' \setminus I). \quad (2.4)$$

Proof. This is [4, Prop. 4.5]: Weyl's inequality (at threshold $\theta \geq \theta_0$, Proposition 2.3) reduces $n_+^\theta(\tilde{G})$ to $n_+(\tilde{A})$, and Proposition 2.2 with the multiplicity count $s_2 + p \leq \frac{1}{2}(N(I') - s_1)$ gives both displays. $\square$

These inequalities hold for any locally finite multiset in the strip invariant under $\rho \mapsto 1 - \bar{\rho}$ with $N(t+1) - N(t) \ll \log(t+3)$; they contain no arithmetic. All the arithmetic is in the trace moments.

2.3 The first two moments, and notation for the ladder

Proposition 2.5 (First two moments). *$\text{tr } \tilde{G} = aL N(T, 2T) + O(L\sqrt{X})$, and*

$$\text{tr } \tilde{G}^2 = 2\pi bL \int_T^{2T} \mu^2 + \frac{T}{\pi} \sum_{n \leq X} \frac{\Lambda(n)^2}{n} g(\log n) + O(Ll \log l (l^2 + X)).$$

Consequently, in the endpoint regime $\lambda \rightarrow 1^-$, $w/L \rightarrow 0$: $\mu_1 \rightarrow 1$ and $\mu_2 \rightarrow \frac{4}{3}$.

Proof. This is [4, Prop. 5.3 and Thm. 5.8]: the trace by the Gabor–Poisson collapse, the second moment by Montgomery's prime-side evaluation of the pair-correlation second moment at bandwidth ≤ 1 , unconditional in the band [12, 1]. We restate it because the diagonal $\sum_{n \leq X} (\Lambda(n)^2/n) g(\log n) = \frac{L^3}{6}(1 + O(1/L))$, with the same window autocorrelation (2.3), is the calibration consumed by the third-moment ledger (Theorem 3.4) and by the model gates of §3.5. $\square$

Remark 2.6 (Endpoint conventions). The endpoint $\lambda = 1$ is admissible because every secondary term above is $o(1)$ relative to the mains even at $X \asymp T$; a reader preferring a power saving throughout may keep $\lambda < 1$ fixed and let $\lambda \rightarrow 1^-$, with the same limits [4, Rem. 6.1]. The taper tail (the $2w/L$ deficit in (2.3)) vanishes as $w/L \rightarrow 0$; all endpoint statements in this paper take the double limit in this order (cf. [4, §7.1]).

By Proposition 2.5 and the ledger of §3, in the endpoint regime,

$$\mu_1 \rightarrow 1, \quad \mu_2 \rightarrow \frac{4}{3}, \quad \mu_3 \rightarrow 2, \quad \mu_4 = \frac{197}{60} + R + o(1), \quad (2.5)$$

where R is the shifted-correlation class (§3.3), of model value exactly $-\frac{1}{30} = \frac{13}{4} - \frac{197}{60}$. The zero-side consumption is the Christoffel-function count of Proposition 2.4: a lower bound on the number of eigenvalues of $\tilde{G}$ exceeding the vanishing threshold θ_0 of Proposition 2.3 converts into the zero counts (2.4), and the mass bound near the origin comes from the moment data

alone (§3). At the moment sequence $(1, 1, \frac{4}{3}, 2, \frac{13}{4})$ the count gives 13/18 and 31/36 exactly; it is monotone in each even moment and decreasing in the odd-moment deficits.

Windows and cells: windows $I_m = [X, 4X]$ and $I_n = (b_2/b_1)I_m$ of length $Y \asymp X$ (more generally $Y \asymp X^\vartheta$, $\vartheta \in (\frac{1}{2}, 1]$); prime-power moduli b_1, b_2 with $g = (b_1, b_2)$, $r = b_1/g$, $q = b_2/g$; shifts $k \leq K \asymp Y/\max(r, q)$ and locks $|j| \leq J \asymp X$; $e(x) = e^{2\pi i x}$; windowed prime sums on the position lattices

$$S_m(\beta) = \sum_{m \in I_m} \Lambda(m) e(b_2 m \beta), \quad S_n(\beta) = \sum_{n \in I_n} \Lambda(n) e(b_1 n \beta).$$

$N_4(k, j)$ denotes the Λ^4 -weighted count of locked quadruples $(m, m - rk, n, n - qk)$ with $b_1 n - b_2 m = j$, and $\mathfrak{S}_4(k, j) V(k, j)$ its singular-series model, where $V(k, j)$ is the window-overlap volume of the cell — the archimedean count of quadruples in the joint window under the lock constraint, of size $\asymp Y^2/(rq\ell_1)$ after consumption normalization — with the conventions of the fourth-moment cell ledger (§4.2), frozen in the reproduction suite. The five-case local table $\kappa_p(v_p(j); v_p(b_1), v_p(b_2))$ defining $\mathfrak{S}_{b_1, b_2}(j) = \prod_p \kappa_p$ is verified digit-exact against residue counting, with $\mathbb{E}_j[\mathfrak{S}] = 1$ (§4.1).

Import status. [4] is a publicly posted preprint; its framework is consumed exactly as stated in this section, and its own analytic imports are the published literature [17, 9, 16, 12, 1]. The averaged shifted-convolution input of the one-sided route is Matomäki–Radziwiłł–Tao [11], used as published. The formalization status of the paper’s layers is graded in §11(f): the identity, local-law and certificate layers are kernel-compiled in core Lean 4; the analytic layer is not formalized.

3 The certificate layer

This section presents the certificate that carries the ladder beyond the rank–trace inequality of [4, Lem. 3.2], and the program’s unconditional moment bookkeeping, machine-verified throughout. It supplies: the Chebyshev–Markov certificate with its exact $\kappa(c)$ laws; the third-moment theorem; the 197/60 decomposition of the fourth moment with the two-bookkeeping reconciliation; and the two-wall structure of the remaining class R , with its audit apparatus.

3.1 The Chebyshev–Markov certificate and the exact exchange rates

Lemma 3.1 (Certificate; machine-verified exactly). *Let ν be a probability measure on $\mathbb{R}$ with moments m_k . Put*

$$Q(x) = \frac{(x^2 - \frac{21}{8}x + \frac{3}{2})^2}{(3/2)^2}.$$

Then $Q \geq 0$, $Q(0) = 1$, $Q \geq 1$ on $(-\infty, 0]$, Q is decreasing on $[0, \frac{21}{16}]$, and

$$\int Q d\nu = \frac{4}{9} \left(m_4 - \frac{47}{16} \right) + R_Q, \quad R_Q := \frac{4}{9} \left[-\frac{21}{4}(m_3 - 2) + \frac{633}{64}(m_2 - \frac{4}{3}) - \frac{63}{8}(m_1 - 1) \right], \quad (3.1)$$

so $|R_Q| \leq 11\varepsilon$ if $|m_1 - 1|, |m_2 - \frac{4}{3}|, |m_3 - 2| \leq \varepsilon \leq 1$. For $0 < \theta \leq \frac{1}{100}$,

$$\nu((-\infty, \theta]) \leq (1 + 4\theta) \left[\frac{4}{9} \left(m_4 - \frac{47}{16} \right) + 11\varepsilon \right].$$

If $m_1 = 1$, $m_2 = \frac{4}{3}$, $m_3 = 2$, $m_4 \leq \frac{13}{4}$, then $\nu((-\infty, \theta]) \leq \frac{5}{36} + O(\theta)$, and this is sharp: the three-atom measure supported on $\{0, a, b\}$, with $a = 0.840635\dots$, $b = 1.784365\dots$ the roots of $x^2 - \frac{21}{8}x + \frac{3}{2}$, attains it. Consequently, under $m_4 \leq \frac{13}{4} + c$ the counts of §2 give the exact linear laws

$$\liminf \frac{N_0^s}{N} \geq \frac{13}{18} - \frac{8c}{9}, \quad \liminf \frac{N_d}{N} \geq \frac{31}{36} - \frac{4c}{9}.$$

Proof. Nonnegativity is clear and $Q(0) = 1$; for $x \leq 0$ the inner quadratic is $\geq \frac{3}{2}$, so $Q \geq 1$; on $[0, a]$ the inner quadratic decreases from $\frac{3}{2}$ through 0 and $a > \frac{1}{100}$; a direct computation gives $Q(\theta)^{-1} \leq 1 + 4\theta$ for $\theta \leq \frac{1}{100}$. Expanding Q and taking expectations,

$$\left(\frac{3}{2}\right)^2 \int Q d\nu = m_4 - \frac{21}{4} m_3 + \left(\frac{441}{64} + 3\right) m_2 - \frac{63}{8} m_1 + \frac{9}{4},$$

and substituting $m = (1, \frac{4}{3}, 2, m_4)$ gives $\frac{4}{9}(m_4 - \frac{47}{16})$; the displayed R_Q collects the first-order deviations, and $\frac{4}{9}(\frac{21}{4} + \frac{633}{64} + \frac{63}{8}) \leq 11$. For sharpness, the three-atom measure with moments $(1, \frac{4}{3}, 2, \frac{13}{4})$ exists by the Hankel recurrence $m_{k+1} = \frac{21}{8}m_k - \frac{3}{2}m_{k-1}$ on the atom set, and Q vanishes doubly at a, b . The linear laws follow by inserting the mass bound into the counts of §2 exactly as in the proof display of (3.1): mass $\leq \frac{5}{36} + \frac{4c}{9} + O(\theta)$, and $1 - 2(\frac{5}{36} + \frac{4c}{9}) = \frac{13}{18} - \frac{8c}{9}$, $1 - (\frac{5}{36} + \frac{4c}{9}) = \frac{31}{36} - \frac{4c}{9}$. $\square$

Remark 3.2. Every identity of Lemma 3.1 is verified symbolically (sympy, exact rationals) in the reproduction package (`certificate_verification.py`) and re-run in the certification audit; the roots, the $5/36$ evaluation, the $\kappa(c)$ laws, and the Hankel floors $m_5^{\text{ext}} = \frac{177}{32} = 5.53125$, $m_6^{\text{ext}} = \frac{2469}{256} = 9.6445\dots$ reproduce exactly. Re-optimizing the certificate polynomial for each c (a linear program) improves the record-breaking threshold from $c < 0.0559$ to $c < 0.0721$ (`moment_lp_reopt.py`: $M^* = 3.3221 = \frac{13}{4} \times 1.0222$); the moment-pinned LP of §9 is the same mechanism run at the full moment sequence.

3.2 The third moment, unconditionally

Lemma 3.3 (Exact 3-cycle resonances). *Let $\lambda \leq 1$ and let $n_1, n_2, n_3 \leq X = T^\lambda$ be prime powers with $|\log(n_1 n_2 / n_3)| \leq C/T$. Then for $T \geq T_0(C)$, $n_1 n_2 = n_3$; the analogous statement holds for every sign pattern of the 3-cycle.*

Proof. $n_1 n_2$ and n_3 are positive integers with ratio $e^{O(1/T)}$; if distinct, their ratio differs from 1 by at least $1/n_3 \geq T^{-\lambda}$, incompatible for large T unless $\lambda = 1$; the borderline is handled by $|n_1 n_2 - n_3| \geq 1$ versus $\leq C n_3 / T \leq C$: for $n_3 \leq T/(2C)$ equality is again forced, and the range $n_3 \in (T/2C, T]$ contributes only through the taper tail, which vanishes in the normalized limit (Remark 2.6). $\square$

Theorem 3.4 (Third moment). *For every fixed $\lambda \in (0, 1]$ and admissible taper, μ_3 converges as $T \rightarrow \infty$, and its limit tends to 2 in the endpoint limit $\lambda \rightarrow 1^-$, $\eta \rightarrow 0^+$.*

Proof (nine-class ledger; full details in the archived memo *V1_completion.md*). Expand $\text{tr } \tilde{G}^3$ against the 3-cycle $\Phi(\tau_1 - \tau_2)\Phi(\tau_2 - \tau_3)\Phi(\tau_3 - \tau_1)$ after the Gabor–Poisson collapse (Lemma 2.1); the tail correction is controlled by the trace–Hölder splitting $\text{tr}(A + E)^3 - \text{tr } A^3$ with $\|E\|_{\text{op}} \leq \theta_0 \ll lT^{\lambda/2-1}$ (Proposition 2.3) and $\|A\|_F^2 = O(d\ell_1^2)$: $o(d\ell_1^3)$ in total. Of the nine classes of $\nu_X^{\otimes 3}$, $\nu_X = \mu + \Pi_X + P_X$: (i) every class containing Π_X is $O(T^{\lambda/2-1})$ pointwise on the window, negligible; (ii) single- P classes are $O(\sqrt{X} l^2 L)$ by $\sum \Lambda(n) n^{-1/2} \leq 3\sqrt{X}$; (iii) the class μ^3 equals

$d\ell_1^3(1 + O(1/l))$ (the ℓ_1 -normalization is exact to $O(1/l)$; measured 1.0048/1.0014/1.0004 at $T = 600/38400/10^8$); (iv) each of the three μP^2 classes reduces, by the $k = 2$ case of Lemma 3.3 plus the Montgomery–Vaughan Hilbert inequality for the kernel tails, to the exact diagonal $\sum_{n \leq X} (\Lambda(n)^2/n) w(\log n)$ with the same $(L - u)_+$ -shaped edge weight as at $k = 2$ (structure constants of the cycle expansion are local to edges and slots, hence pinned by the $k = 2$ case, where the total is Montgomery’s evaluation), and $\sum_{n \leq X} (\Lambda^2/n)(L - \log n) = L^3/6(1 + O(1/L))$: each class contributes $\frac{1}{3}d\ell_1^3(1 + o(1))$ at the endpoint, the three positions summing to 1; (v) the class P^3 is forced by Lemma 3.3 onto exact relations $n_1 n_2 = n_3$, i.e. powers of a single prime, and $\sum_p \sum_{a+b=c} \log^3 p p^{-c} = O(1)$ makes it $O(Tl)$. Summing: $\mu_3 \rightarrow 1 + 1 = 2$. Cross-checks: the $k = 2$ instance of the same constant chain reproduces Montgomery’s $\frac{1}{3}$ to four digits; the model gate $\int O_3 C_3 = 0$ is exact (§3.5); the finite-height shape $m_3 m_1 / m_2^2$ matches the model to 2%. Two nonstandard steps — the $k = 3$ tail adaptation of Proposition 2.3 and the locality-plus-calibration fixing of the constant chain — are written out in the archived memo and are consumed at that rigor. $\square$

3.3 The fourth moment: the 197/60 ledger and the class R

Proposition 3.5 (Evaluable classes of the fourth moment). *By the identical ledger at $k = 4$ (81 classes; Π_X - and single- P classes negligible; paired sign patterns forced exact by Lemma 3.3; $\{2, 2\}$ -pairing weights explicit window integrals),*

$$\mu_4 = \underbrace{1}_{\mu^4} + \underbrace{2}_{\mu^2 P^2 \text{ diag}} + \underbrace{\frac{17}{60}}_{\{2,2\} \text{ bare pairings}} + R + o(1) = \frac{197}{60} + R + o(1)$$

in the endpoint regime, where $\frac{17}{60} = 2\kappa$ with $\kappa = \int_{[0,1]^2} xy \Sigma_{\text{pair}} O_4 = \frac{17}{120}$ (exact quadrature of the certified 4-cycle overlap geometry), and R is the shifted-correlation class: pairs of restricted products $N_1 = b_1 m_1 \neq N_2 = b_2 m_2$ (all legs prime powers $\leq X$), weight $\Lambda^4 / \sqrt{N_1 N_2}$, window kernel $\widehat{W}(\Delta) = [\sin 2T\Delta - \sin T\Delta] / \Delta$ at $\Delta = \log N_1 / N_2$, and the explicit overlap geometry O_4 (row/column asymmetric; unordered-cell weight $Q(\beta_1, \beta_2) + Q(\beta_2, \beta_1)$); R -units $= (1/\pi) \Sigma / (d\ell_1^4)$. The model value of R is exactly $-\frac{1}{30} = \frac{13}{4} - \frac{197}{60}$.

Remark 3.6 (Bookkeeping reconciliation; audit rounds 25–26). Two bookkeeping conventions coexist in the program archive and must not be mixed. The *bare-diagonal* convention above books the $\{2, 2\}$ pairings at $\frac{17}{60}$ and carries the pair-level Montgomery plateau (the $\min(u, 1)$ saturation of C_2) inside R . The *sine-model cumulant* convention books $\{2, 2\}$ as $2t_{\text{adj}} + t_{\text{opp}}$ with the connected part separate. With the anchors now certified as exact rationals (§8.3: $t_{\text{adj}} = \frac{7}{60}$, $t_{\text{opp}} = \frac{1}{30}$), the reconciliation is exact and symmetric:

$$\underbrace{\frac{17}{60}}_{\text{bare}} = \underbrace{\frac{4}{15}}_{2t_{\text{adj}} + t_{\text{opp}}} + \underbrace{\frac{1}{60}}_{\text{plateau}}, \quad R_{\text{model}} = \underbrace{-\frac{1}{60}}_{\text{plateau (pair-of-pairs Wick)}} + \underbrace{-\frac{1}{60}}_{\text{connected core}} = -\frac{1}{30},$$

and $1 + 2 + \frac{17}{60} = \frac{197}{60}$, $2 \cdot \frac{17}{120} = \frac{17}{60}$ exactly. The archived grid values (0.2677, plateau 0.0156, connected -0.0177 ; cross-convention measurement -0.01567 in `m1_suite.py`) carried a small grid bias in t_{adj} (quoted 0.11717, exact $\frac{7}{60} = 0.11667$; register item D7); the exact connected model value $-\frac{1}{60} = -0.01667$ agrees with the cumulant engine’s independent grid measurement -0.01663 to 0.2%. The totals consumed anywhere in this paper were and remain exact ($\frac{197}{60}$, $-\frac{1}{30}$, $\frac{13}{4}$); the split is now exact too. All ledgers below state which convention they use.

3.4 The two walls of R , and its direct measurement

Everything in this subsection is unconditional structure or finite-height measurement; it is the audit apparatus against which the proofs of §§5–7 were developed and checked. (i) *Two ghost lemmas*. The mean-field quadruple term is $O(l^{-3})$: the u -Jacobian cancels $1/\sqrt{N_1 N_2}$ exactly and the inner kernel integral is $-2[\text{Si}(2T/N) - \text{Si}(T/N)]$, a boundary layer at $N \asymp T$; and the smooth part of the cluster-squared sector vanishes identically, since $\int_{\mathbb{R}} \widehat{W}_{\text{in}} d\Delta = 0$. (ii) *Two walls*. Consequently $R = R_{\text{pp}} + R_{\text{conn}}$: an edge layer carried by Poisson-dual lattice terms at top scale (some leg within $O(1)$ of X ; the explicit-formula regime) and a bulk object (model bands at $s \sim 1.5$). (iii) *Separation observable, certified at three heights*: binning by $e = l - \max u_i$ gives, at $l = 5.95/6.64/7.33$: edge $-0.0061/-0.0068/-0.0060$ (saturating toward the model -0.0156) and bulk $+0.0004/-0.0012/-0.0015$ (sign turn at $l \approx 6$, monotone toward -0.0177). (iv) *Direct measurement*. Meet-in-middle enumeration of all resonant quartic prime pairs (exact kernels, class-resolved, flat window, $\lambda = 1$) gives

l	4.56	5.94	7.33	8.72
R (per $d\ell_1^4$)	-0.0008	-0.0057	-0.0074	-0.0101

— negative at every height, trending at the $1/l$ -rate of all calibration gates; a two-code-path audit showed the sharp cutoff $C = 40$ understates $|R|$ (full-kernel -0.0012 and -0.0098 at the first two heights), so these are magnitude lower bounds. The same audit certified the ledger’s constant chain end-to-end at $k = 2, 3, 4$ to 0.05–0.8%. The $k=1$ Poisson stationary term of the edge layer is evaluated in closed form (zone-1 master formula; validated at three heights and two tapers to 5–18%): a positive decaying transient $\sim cl/\ell_1^4$ explaining the late sign turn of the near band.

3.5 Model calibration

The free-fermion cumulant engine computes the Fourier-field cumulants of the sine process by the ordered-set-partition expansion of $\text{Tr} \log(1 + (e^f - 1)K)$; gates: $C_2(v) = \min(|v|, 1)$ exactly, $\int O_3 C_3 = 0$ (whence $m_3(1) = 2$), and the $m_4(1)$ assembly 3.25103 against $\frac{13}{4}$ (grid error 0.03%). The value $m_4(1) = \frac{13}{4}$ is pinned three independent ways (cumulant assembly; GUE simulation $m_{1.4} = 1.0000, 1.3313, 1.9913, 3.2213(27)$; Hankel forcing from the extremal constants). GUE double-window Richardson extrapolation measures the model values

$$m_5(1) = 5.573 \pm 0.113, \quad m_6(1) = 10.02 \pm 0.27, \quad (3.2)$$

(sine-model/GUE measurement — see the discrepancy register, Appendix B, item D2, for the provenance correction of an earlier attribution), and the six-moment certificate at the central values gives 0.7328/0.8664: the ladder climbs only through the odd moments’ strict-positivity gaps, since the $k \leq 4$ extremal already has $m_5 = \frac{177}{32}$, $m_6 = 9.6445$. The assembled values of §8 lie inside the bands (3.2).

4 The one-sided fourth-moment route: 0.6916

This section absorbs the fourth-moment extension paper (v4): a one-sided bound on R sufficient to cross the pair-correlation ceiling 0.68185, by classical technology plus the averaged shifted-convolution theorem of [11]. It is logically independent of the exact evaluation of §§5–7 and is

retained both as the program's first crossing of the ceiling and because its architecture (cells, bridge, aggregation, discharge ledger) is consumed by the Lemma D proof.

Theorem 4.1 (One-sided fourth-moment bound; grade: certified-candidate with the numerical-analysis step of Lemma 4.10). *With $R = R(1)$ the class of Proposition 3.5, $\limsup_{T \rightarrow \infty} R(1) \leq \bar{R} := 0.0066$. Consequently, by the count of §2 at $(1, 1, \frac{4}{3}, 2, \frac{13}{4} + \Delta)$, $\Delta = \bar{R} + \frac{1}{30} = 0.039933$,*

$$\liminf \frac{N_0^s}{N} \geq 1 - 2\Lambda_2(0) = 0.691615, \quad \liminf \frac{N_d}{N} \geq 1 - \Lambda_2(0) = 0.845807,$$

where $\Lambda_2(0) = \frac{5/108 + \Delta/3}{1/3 + 4\Delta/3} = 0.154193$, and the same for any fixed primitive Dirichlet L -function. Any $\bar{R} \leq 0.0222$ crosses the 0.68185 ceiling; the margin is 0.0156 in R -units.

(An earlier draft printed $\Lambda_2(0) = 0.156293$; the certification audit recomputed the closed form and found the stated theorem constants 0.6916/0.8458 correct and that figure stale — discrepancy register D1, Appendix B.) The constant $\bar{R}$ decomposes as $[\text{core} \leq -0.0055] + [\varepsilon_{\text{tail}} \leq 0.0111] + [\varepsilon_{CL} = 0.0010]$, where ε_{CL} is a convergence-slack allowance for the core sequence (Lemma 4.10).

4.1 Toolkit: singular series, arc identity, tails

For $b_1, b_2 \geq 1$, $j \neq 0$ the local density of $b_1 m_1 - b_2 m_2 = j$ is $\mathfrak{S}_{b_1, b_2}(j) = \prod_p \kappa_p$ with the five-case table verified digit-exact; $\mathbb{E}_j[\mathfrak{S}] = 1$. Its arc expansion:

Proposition 4.2 (Arc identity). $\mathfrak{S}_{b_1, b_2}(j) = \sum_{q \geq 1} \beta_q c_q(j)$, $\beta_q = \frac{\mu(q_1)\mu(q_2)}{\varphi(q_1)\varphi(q_2)}$, $q_i = \frac{q}{(q, b_i)}$.

At $b_1 = b_2 = 1$ this is the classical evaluation used in [11, (66)–(67)]; the general case follows by the same p -local comparison against the κ_p table. The truncation tail obeys, for $g = (b_1, b_2) \leq (\log M)^C$,

$$\sum_{|j| \leq H} (\mathfrak{S} - \mathfrak{S}^{(P)})^2(j) \ll H (\log)^{-B+O(1)}, \quad P = (\log)^B, \quad (4.1)$$

by the (q_1, q_2) -grading and $\sum_{|j| \leq H} c_q(j)^2 \ll (H + q)\varphi(q)$. *Discharge ledger* (kept, per the program's audit discipline): cells with $g > (\log)^C$ are discarded at certified cost 0.71% weight share, $\leq 3 \times 10^{-4}$ R -units. The trivial envelope $\sum_{n \leq x} \Lambda(n)\Lambda(n+h) \ll \mathfrak{S}(h)x$ uniformly for $|h| \leq x$ is [8, Cor. 3.14].

4.2 Covered zone, middle band, bridge and aggregation

Write $S_i(\alpha) = \sum_{m \sim M_i} \Lambda(m)e(b_i m \alpha)$. The two-modulus structure is discharged by four archive lemmas (rounds 38–40): the major-arc law $S_i(\alpha) = \frac{\mu(q_i)}{\varphi(q_i)} v_i(\eta) + O(M_i \exp(-c\sqrt{\log M}))$ with $q_i = q/(q, b_i)$ (b -uniformity free); the classification of joint near-major points ($q \leq \min(P^2 g, P b_1, P b_2)$), a finite b -independent family whose content is the $(q_1, q_2 \leq P)$ -graded truncation $\mathfrak{S}^{(P)}$; the dilation transfer ($\gamma_i = b_i \alpha$ obeys Vaughan's estimate in *its own* approximation quality); and a smoothing-collar lemma aligning the Gallagher width. After this discharge the $b = 1$ machinery of [11] applies verbatim:

Proposition 4.3 ([11, Thm. 1.3(i)], variance form). *For $X^{8/33+\varepsilon} \leq H \leq X^{1-\varepsilon}$, any center $h_0 \leq X^{1-\varepsilon}$, and any A :*

$$\sum_{|h-h_0| \leq H} \left| \sum_{X < n \leq 2X} \Lambda(n)\Lambda(n+h) - \mathfrak{S}(h)X \right|^2 \ll_{A,\varepsilon} HX^2(\log X)^{-A}.$$

The covered share of the class at the 8/33 exponent is 0.885 (anchored at the archived 0.797 at exponent $\frac{1}{3}$). For the middle band, with $A(b_2, j) = \sum_m \Lambda(m)\Lambda((b_2m+j)/b_1)\mathbf{1}[b_1 \mid b_2m+j]$ and $MT = \mathfrak{S}_{b_1, b_2}(j) \text{len}/(b_1b_2)$:

Lemma 4.4 (Dispersion swaps; exact). *$\sum_{b_2, j} A^2$, $\sum A \cdot MT$ and $\sum MT^2$ are, exactly, weighted twin-sum families over the structured shifts $h = qk$ (outer pairs $m - m' = rk$), the $\mathfrak{S}$ -against- Λ correlation on the strip, and $2J E_2(b_1, b_2)(\text{len}/b_1b_2)^2(1+o(1))$ respectively, where $E_2 = \prod_p \mathbb{E}_v[\kappa_p^2]$ (anchor: $E_2(1, 1) = 2 \prod_{p>2} (1 + (p-1)^{-3})$).*

Lemma 4.5 (Bridge). *For any window W and modulus q : $\sum_{k \neq 0} \sum_{n, n-qk \in W} \Lambda(n)\Lambda(n-qk) = \sum_{a \bmod q} \left[\psi_W(q, a)^2 - \sum_{n \in W, n \equiv a} \Lambda(n)^2 \right]$.*

Lemma 4.6 (Structured-to-full aggregation). *Let W have length Y , $\text{dev}(h)$ the (smoothly weighted) twin deviation on W , Q a set of moduli with shift counts $K_q \leq H/q$, $H \leq Y^{1-\varepsilon}$. If $\sum_{h \leq H} |\text{dev}|^2 \ll_{A'} HY^2(\log)^{-A'}$ for every A' , then for every A*

$$\sum_{q \in Q} \sum_{k \leq K_q} |\text{dev}(qk)|^2 \ll_A HY^2(\log Y)^{-A},$$

uniformly in Q — no factor of the moduli is lost.

Proof. The left side is $\sum_{h \leq H} \nu(h) |\text{dev}(h)|^2$ with $\nu(h) \leq \tau(h)$. Split at $\nu \leq (\log Y)^C$: the generic part costs $(\log)^C$ against the arbitrary- A' input; on the rare part use the envelope $|\text{dev}(h)| \leq 9\mathfrak{S}(h)Y$ with $\sum_{h \leq H, \tau(h) > V} \tau(h) \ll H(\log)^3/V$ at $V = (\log)^C$. Choose $C = A + 4$, $A' = 2A + 4$. $\square$

The band consumer applies a single Cauchy–Schwarz *in the shift variable only* (any Cauchy–Schwarz across the family pays its Poisson floor, measured 50–60 $\times$ over budget, and is never used). The glue layer (welding weight $w_k(n) = \sum_{m \in I(n)} \Lambda(m)\Lambda(m-rk)$) closes by the exact factorization of main terms, Abel summation against the glue for the major arcs, and the divisor-bounded envelopes of [11, Prop. 5.4, App. A] for the minor arcs.

4.3 Deterministic mains and the continuum limit

Write $\mathfrak{S}_{b_1, b_2} = \bar{D} + g_P + g_{\text{tail}}$ ($\bar{D}$ the exact finite local mean; g_P the $q \leq P$ oscillating piece; $P = 40$).

Lemma 4.7 (Partial-sum identity). *For every $M \geq 1$, $\sum_{j \leq M} (\mathfrak{S}_{b_1, b_2} - \bar{D})(j) = - \sum_{d \geq 1} d \left\{ \frac{M}{d} \right\} \gamma_d^{\text{free}}$,*

with explicit Euler-local γ , absolutely convergent against the ℓ^1 weights $d \min(1, M/d)$.

Lemma 4.8 (Finite- J evaluation and uniform tail). *For every finite J and θ , $\sum_{j \leq J} \text{tail}_P(j) \text{row}_j(\theta) = \sum_{d \geq 1} \gamma_d^{>P} S_0(d\theta; \lfloor J/d \rfloor)$ with $S_0(x; K) = \sum_{k \leq K} \frac{\sin 2xk - \sin xk}{k}$; hence the θ -uniform bound $\leq C_{\text{saw}} \sum_d |\gamma_d^{>P}|$. No $J \rightarrow \infty$ interchange is performed anywhere.*

Lemma 4.9 (Certified sawtooth constant). $|S_0(x; K)| \leq C_{\text{saw}} := 2.10$ for all x and $K \geq 1$.

Proof. For $K \leq 64$: rigorous grid verification with the Lipschitz bound $|\partial_x S_0| \leq 3K$ certifies $\max |S_0| \leq 1.8675$. For $K > 64$: the Dirichlet representation with the sine-integral comparison near the singular set and the second mean value theorem away from it gives ≤ 2.05 ; samples at $K = 128, 512, 4096$ read 1.842, 1.849, 1.851, approaching $\text{Si}(\pi) = 1.85194$ from below. (Re-run in the certification audit: identical values.) $\square$

Zone-integrating the uniform tail against the certified geometry gives $\varepsilon_{\text{tail}} \leq 0.0115$ ($T = 2400$), 0.0111 ($T = 4800$), decreasing; the evaluated core (exact finite forms) is $\text{core}(2400) = -0.00549$, $\text{core}(4800) = -0.0064$, certified against the pipeline anchor (-0.00545 vs the suite's -0.00544).

Lemma 4.10 (Continuum limit of the core; grade: proof shape with computed constants — see the grading note below). *$\text{core}(T)$ converges as $T \rightarrow \infty$ to $C_{\text{core}} = \sum_{b_1 \leq b_2} \frac{\Lambda(b_1)\Lambda(b_2)}{b_1 b_2} F_{\infty}(b_1, b_2)$, an absolutely convergent series of computable continuum sawtooth integrals, with $|\text{core}(T) - C_{\text{core}}| \ll (\log T)^{-1}$; the ensemble collapses onto the universal $(1, 1)$ -class (Mertens mass concentrates on pairs of large primes), whose slope and constant are computed, not fitted: partial-sum slopes $0.99684 \rightarrow 0.99962 \rightarrow 0.99996$ over successive decades (classical law, exactly 1 — now proved: the Dirichlet series is $\zeta(s+1)E(s)$ with $2C_2E(0) = 1$, Proposition C.1), and the constant $c_0 = -1.4228$ is now identified in closed form: $c_0 = \gamma - 2 = -1.4227843 \dots$ (Euler–Mascheroni γ ; Proposition C.1, verified to 10^{-5} at $M = 2 \times 10^7$); θ -ladder $c^* = -0.5433$, decade-stable, matching the archive's independent $-0.55(2)$ ($c^* = \log \pi + \gamma - 2 - \overline{\text{osc}}$ in the pin of the archived F -QC5 gate, so the remaining computed piece of c^* is the mean oscillation $\overline{\text{osc}} \approx 0.2652$, not identified here). The continuum quadrature with Richardson extrapolation in $1/l$ ($l = 40, 60, 100, 160$) yields $C_{\text{core}} = -0.0209 \pm 0.0026$, independently confirming the archive ladder $-0.020(2)$ — two disjoint routes at 4%.*

Grading note. The convergence structure (scaled-variable Riemann sum, universality collapse) is an analytic argument; the values of the collapsed series' slope and constant, and the quadrature of C_{core} , are computed constants with stated bands, not certified enclosures. Theorem 4.1 charges only the conservative core $\leq -0.0055 + \varepsilon_{CL}$, $\varepsilon_{CL} = 0.0010$, against the measured decreasing grid; the certification write-out of the quadrature is a named open item (§11). Substituting the computed C_{core} would give $\bar{R} = -0.0098 \pm 0.0026 < 0$, hence 0.7031/0.8515; that upgrade is *not* consumed anywhere in this paper.

4.4 Assembly

At $\lambda = 1$, uniform full-kernel convention:

$$R(1) \leq \text{core}(T) + \varepsilon_{\text{tail}} + o(1) \leq (-0.0055 + 0.0010) + 0.0111 + o(1) = 0.0066 + o(1),$$

the $o(1)$ collecting the covered-zone, band and low-zone deviations of §4.2 (the low-zone cap's certified finite-height price at the operating point is 0.0163, inside the LP budget 0.0387+0.0177). Feeding $\bar{R} = 0.0066$ into the count gives Theorem 4.1. $\square$

5 The truncation discharge

Lemma 5.1 (Cell-mass bound). *Uniformly in the lock and shift ranges, the Λ^{2k} -weighted, consumption-normalized mass of the k -cycle cells at parameter ℓ_1 is $\ll Y^2(\log Y)^c \ell_1$.*

Proof. For a fixed cell, each locked configuration is determined by (m, k) together with the residue data of the lock; the count of (m, k) pairs in the windows is $\ll YK \ll Y^2/\max(r, q)$, each carrying weight $\ll (\log Y)^{2k}$, and the lock constraint selects a $1/\ell_1$ -proportion of the n -range up to divisor-bounded multiplicity; the consumption normalization (division by the ledger volumes $V \asymp Y^2/(rq\ell_1)$) leaves at most one factor ℓ_1 . The numerical ledger face measures the aggregate mass essentially flat in ℓ_1 (fitted exponent -0.11), well inside the bound. $\square$

Lemma 5.2 (Multiplicity). *The number of unordered prime-power tuples with cell parameter $\ell_1 = \ell$ is $\ll_\varepsilon \ell^\varepsilon$ (divisor-bounded).*

Proposition 5.3 (Truncation–consumption compatibility). *Let F denote the consumed k -cycle cell aggregate, $k \geq 4$, and $F^{(P)}$ its restriction to $\ell_1 \leq P$. Then*

$$|F - F^{(P)}| \ll_\varepsilon Y^2(\log Y)^c P^{-(k-2)+\varepsilon} \quad \text{two-sidedly.}$$

Hence for every A_0 there is B (any fixed $B > (c + A_0)/(k - 2)$ suffices) such that proving the variance bound for moduli $\leq P = (\log X)^B$ proves it in the consumed form, with total truncation cost $\ll (\log X)^{-A_0}$: only moduli $\leq (\log X)^B$ are ever consumed.

Proof. Weight $\times$ mass $\times$ multiplicity $= \ell^{-k} \cdot \ell \cdot \ell^\varepsilon$ summed over $\ell > P$. The bound is on absolute values, hence two-sided. The archived D1 layer independently certifies the squared-remainder grading at $k = 4$ ($\mathbb{E}_j[(F - F^{(P)})^2] \ll P^{-1-\delta}$; tail locals $t_p^2 \sim 4/p^2$). Two truncations coexist in the program and must not be confused: the sawtooth/frequency truncation inside the C_{full} -quadrature machinery of §4.3, whose tail carries that limit, and the present ℓ_1 -cell truncation, whose tail is trivially small; the audit trail (archive rounds 91–93, item S1) verified the disentanglement. An earlier draft stated the multiplicity as ℓ_1 ; the divisor bound of Lemma 5.2 is what the displayed exponent uses. $\square$

6 The local closure

The local (mod p) structure of every moment order is governed by affine lock chains. Fix a cycle length $b \geq 2$, a prime power $P_0 = p^s$, unit coefficients $a_1, \dots, a_b$, and free locks $j_1, \dots, j_{b-1} \in \mathbb{Z}/P_0$; define points by $n_{t+1} \equiv a_t n_t + j_t$, the last lock induced by cyclic closure. Write $n_t = A_t(n_1 + d_t)$ where $A_{t+1} = a_t A_t$, $A_1 = 1$, $d_1 = 0$; then $d_{t+1} - d_t = j_t/A_{t+1}$ is a unit multiple of j_t . Let $N(j)$ count the $n_1 \in \mathbb{Z}/P_0$ with all b points prime to p .

Proposition 6.1 (Coincidence-counting closure). *Pointwise in the lock vector j ,*

$$N(j) = P_0 \frac{p - D(j)}{p}, \quad D(j) = \#\{d_t \bmod p : t = 1, \dots, b\},$$

and for any class $\mathcal{C}$ of locks,

$$\kappa_b(\mathcal{C}) = \frac{1 - \mathbb{E}[D \mid \mathcal{C}]/p}{(1 - 1/p)^b}, \quad \mathbb{E}_j[\kappa_b] = 1 \text{ exactly.}$$

Forbidden-class merges are pair events of weight $1/(p - 1)$; triple merges enter at $O(p^{-2})$.

Proof. Since A_t is a unit, $n_t = A_t(n_1 + d_t)$ is prime to p iff $n_1 \not\equiv -d_t \pmod{p}$; the all-unit condition excludes exactly the $D(j)$ distinct residues $\{-d_t\}$, each of density $1/p$ in $\mathbb{Z}/P_0$, giving the count. The class formula is its average against $(1 - 1/p)^b$. For the mean-one identity: under uniform locks the increments are independent uniform, so $(d_2, \dots, d_b)$ is uniform on $(\mathbb{Z}/p)^{b-1}$ and $\mathbb{E}[D] = p(1 - (1 - 1/p)^b)$, whence $1 - \mathbb{E}[D]/p = (1 - 1/p)^b$. A merge across a gap of lock-sum σ requires $p \mid \sigma$: impossible for a single unit lock; for a sum of two units it occurs for exactly one value of the second, weight $1/(p - 1)$. $\square$

Machine verification (pipeline, `face_local_closure`; re-run in the certification audit with identical output): zero pointwise violations over full enumerations and closed-form match to 2×10^{-16} for every lock class at $b = 4, 5, 6$, $p \in \{5, 7, 11, 13\}$, $s \in \{1, 2\}$, with the mean-one identity exact. The κ_4 closed forms of the fourth-moment table are instances ($\kappa_4(0, 0) = 1 - (4 - \frac{2}{p-1})/p$). The proposition supplies the singular-series combs at every cycle length and the j -averaged factorization identities used below. (A first draft of the corresponding Lean seed omitted the normalising inverse in $d_t = B_t A_t^{-1}$; the exact-enumeration proxy fired — registry r103, the 29th conversion — and the seed was corrected: Appendix B.)

7 Lemma D and the exact fourth moment

The fourth-moment evaluation $m_4 \rightarrow 13/4$ closes the estimate isolated by the program (“Lemma D”: the averaged, lock-resolved quadruple-correlation variance) in its consumed, truncated form. In its *unrestricted* form — moduli up to a power of X — it remains open, one structural step beyond the published diagonal cases of the Barban–Davenport–Halberstam/dispersion lineage. Two observations remove the difficulty from the consumed form: the truncation discharge of §5, and the spectral frame below.

7.1 The frame and the Parseval identity

By the archived, machine-verified bridge and glue identities (§4.2), the (k, j) -aggregated variance for a modulus pair is carried by the *dilated density* $\rho(\tau)$: with A_m, A_n the autocorrelations of the two position-lattice sequences, on the analysis grid of period N (a power of two exceeding four times the largest position, so no correlation wraps),

$$\rho(\tau) = \sum_t A_m(t) A_n(t - \tau) = \text{irfft}(|S_m|^2 |S_n|^2)(\tau),$$

and the lock-variance obeys the *exact* discrete Parseval identity

$$\sum_{\tau=0}^{N-1} \rho(\tau)^2 = \frac{1}{N} \left[\text{dens}_0^2 + 2 \sum_{0 < \beta < N/2} \text{dens}_\beta^2 + \text{dens}_{N/2}^2 \right], \quad \text{dens} = |S_m|^2 |S_n|^2, \quad (7.1)$$

verified bitwise in the pipeline (10^{-16} relative; re-run in the certification audit). The same identity applied to ρ –Model turns the variance into the spectral L^2 -distance of dens from the model comb. We classify the frequencies β by the arc quality of the pair $(\gamma_m, \gamma_n) = (b_2\beta, b_1\beta) \bmod 1$ at cutoff $P^* = (\log X)^{B^*}$: both major (MM), mixed, or both minor.

7.2 The three leaves

Lemma 7.1 (MM leaf). *Let β be a doubly major bin: $\gamma_m = a/Q + \eta$, $Q \leq P^*$, $|\eta| \leq \text{arc width}$, and likewise γ_n . Then, for every A' ,*

$$S_m(\beta) = \frac{\mu(Q)}{\varphi(Q)} I_m(\eta) + O_{A', B^*}(Y(\log Y)^{-A'}), \quad I_m(\eta) = \sum_{m \in I_m} e(m\eta),$$

uniformly, and the same for S_n ; the constants are ineffective. Consequently, on the MM bins, $|\text{dens} - \text{comb}| \ll \text{main} \cdot Y(\log Y)^{-A'} + Y^2(\log Y)^{-2A'}$ pointwise, where comb is the product of the squared main terms.

Proof. $S_m(\beta) = \sum_m \Lambda(m) e(m\gamma_m)$ is a windowed single prime sum at $\gamma_m = a/Q + \eta$. Splitting into progressions mod Q and applying the Siegel–Walfisz theorem in the form [6, §22] (or [9, Cor. 5.29]) with partial summation in the window against the phase $e(m\eta)$ gives the display; the constants depend only on (A', B^*) and are ineffective through Siegel’s theorem. The pointwise comb comparison follows by expanding the product. The per-bin face (`face_sw`) measures the absolutely normalised error across six families: RMS 3.6×10^{-3} at $T = 2400$ falling to 3.6×10^{-4} at $T = 153600$. $\square$

Remark 7.2 (Ineffectivity enters here). This is the single point where ineffective constants enter the chain: Siegel’s theorem gives no computable $T_0(A')$, so every theorem constant downstream is ineffective, while every finite-height face measured in §10 carries effective constants. The two kinds of statement are graded separately throughout (see the closing remark of §10); nothing else in the paper — the identities, the certificate, the local law, the window calculus — consumes an ineffective input.

Lemma 7.3 (Minor and mixed leaves). *For γ minor at cutoff P^* (denominator range $((\log X)^{B^*}, Y(\log X)^{-B^*})$ by Dirichlet approximation), $|S(\gamma)|^2 \ll Y^2(\log Y)^{-B^*/2+c_0}$ by Vaughan’s estimate. Hence the doubly-minor spectral mass carries a $(\log)^{-B^*+2c_0}$ saving squared, and each mixed bin at least the one-sided saving; the mixed and minor contributions to the spectral L^2 -distance are $\ll (\log X)^{-B^*/2+c_1}$ relative to the natural size. The size bookkeeping (well-spacing of image families and the hybrid large-sieve/minor-sup stratum budget) is classical and is face-checked at $24\text{--}31\times$ tightness (`face_arcs`).*

Lemma 7.4 (The comb is the singular series). *The spectral main comb of Lemma 7.1 equals the truncated singular-series model: the surviving frequencies are exactly the CRT progression $t \equiv 0 \pmod{b_2}$, $t \equiv \tau \pmod{b_1}$ in the τ -frame, and the local factors match the coincidence-counting closed form of Proposition 6.1; the truncation tail of the singular series beyond P^* is $\ll (\log X)^{-B^*+c}$ by the standard $\sum_{q > P^*} \mu^2(q)(q, h)/\varphi(q)^2$ bound. The identification is displayed explicitly, with a worked modulus pair, in Appendix A; the τ -resolved model tracks the measured profile at 0.2–4% across sixteen families and eight heights (`face_dispersion`).*

7.3 The theorem

Theorem 7.5 (Lemma D, truncated form). *Let B be fixed, $b_1, b_2 \leq (\log X)^B$ prime powers, windows of length $Y \asymp X^\vartheta$, $\vartheta \in (\frac{1}{2}, 1]$. Then for every $A > 0$,*

$$\sum_{k \leq K} \sum_{|j| \leq J} |N_4(k, j) - \mathfrak{S}_4(k, j)V(k, j)|^2 \ll_{A, B} KJV^2(\log X)^{-A}.$$

Consequently, with Proposition 5.3, $m_4 \rightarrow 13/4$ two-sidedly.

Proof. Choose $B^* = 2(A + c_1 + B)$ and apply (7.1) to ρ -Model. The spectral L^2 -distance splits over the arc classification. On MM bins, Lemma 7.1 with $A' = A + 2B^*$ gives a pointwise relative saving $(\log)^{-A'}$, hence $\sum_{\text{MM}} |\text{dens} - \text{comb}|^2 \ll (\log)^{-2A'+c} \sum \text{main}^2$; by Lemma 7.4 the comb agrees with the model to $(\log)^{-B^*+c}$ relative, absorbed by the choice of B^* . The mixed and doubly-minor bins contribute $\ll (\log)^{-B^*/2+c_1}$ relative by Lemma 7.3. The (k, j) -aggregation constants of the bridge and glue layer are archived and certified (§4.2); the normalization match to KJV^2 is the consumer interface there. All bounds are on absolute values, hence two-sided; all constants depend only on (A, B) and are ineffective. Choosing B^* then A' in that order involves no circularity. $\square$

Remark 7.6 (The progression-frame route; falsifier r95). The archived reduction of the variance to a locked two-modulus progression covariance suggests a shorter argument: expand both progression errors in additive characters and evaluate by Siegel–Walfisz. Executed at the moduli (r, q) separately this produces *incomplete* character sums (the lock couples residues modulo r to residues modulo q , and the a -sum over one modulus has no orthogonality against the other); the expansion must be lifted to the joint modulus $M \asymp rq \leq (\log X)^{2B}$, where it becomes a complete-sum argument parallel to the one above. An earlier draft presented that sketch without the lift; it was replaced by the spectral proof, in which the τ -transform runs over the full period and the orthogonality (Lemma 7.4) is exact. The slip and its correction are falsifier r95 on the forward record.

Corollary 7.7 (The 13/18 rung). *With ineffective constants,*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{13}{18}, \quad \liminf_{T \rightarrow \infty} \frac{N_d(T, 2T)}{N(T, 2T)} \geq \frac{31}{36},$$

and the same for any fixed primitive Dirichlet L -function.

Proof. Theorem 7.5 and Proposition 5.3 give $m_4 = 13/4 + o(1)$ two-sidedly; Theorem 3.4 gives $\mu_3 \rightarrow 2$; the certificate (Lemma 3.1, equivalently the Christoffel-function count of Proposition 2.4 at $(1, 1, \frac{4}{3}, 2, \frac{13}{4})$) gives 13/18 and 31/36 exactly, monotonicity in the fourth moment converting $m_4 \rightarrow 13/4$ into the two liminf bounds. The Dirichlet L -function case is as in [4, Thm. E]: the framework uses only the functional-equation symmetry $\rho \mapsto 1 - \bar{\rho}$, the zero-density bound, and the explicit formula, all available for any fixed primitive Dirichlet L -function. Constants are ineffective through Lemma 7.1. $\square$

8 The fifth and sixth moments

8.1 The product identities

Proposition 8.1. *For a b -cycle of position lattices sharing a window, with any per-lattice densities f_i, g_i : $\sum_j N_f(j)N_g(j) = \sum_t \prod_i C_i(t)$, $C_i(t) = \sum_x f_i(x)g_i(x+t)$. In particular the lock variance is the pointwise product of the lattice autocorrelations, summed over the common shift.*

Proof. Two chains share a lock vector iff they differ by a common shift; the chain sum then factorizes over the positions. $\square$

This identity ($O(N \log N)$ to evaluate) transports the spectral argument of §7 *factor by factor* to every cycle length: each factor is a windowed single prime sum, so the three-leaf closure

(per-factor Siegel–Walfisz on all-major bins, Vaughan on any minor factor, comb by Proposition 6.1, truncation by Proposition 5.3 with stronger exponents) applies verbatim. Machine faces: the twin-singular-series product model tracks the empirical five-fold product at 0.28–2.5% (six modulus patterns, $T = 4800$ to 76800 , improving with height) and the six-fold product at 0.11–1.29%.

8.2 Pair layers collapse onto certified anchors

The $\text{Bell}(5) = 52$ and $\text{Bell}(6) = 203$ class ledgers are machine-enumerated (class tables $1/10/15/10/10/5/1$ and $1/15/45/15/20/60/10/15/15/6/1$; pentagon chords $10 + 5$; hexagon chords $30 + 15$; matching crossing spectrum $5/6/3/1$). Frozen-slot identities — a singleton block repeats a walk value and leaves the overlap range unchanged; machine-exact — reduce every pair class to the four-cycle geometry, whose anchors are now *exact rationals* (§8.3): $t_{\text{adj}} = \frac{7}{60}$, $t_{\text{opp}} = \frac{1}{30}$, $\Phi_4 = \frac{1}{4} - 2t_{\text{adj}} - t_{\text{opp}} = -\frac{1}{60}$ (sine-model cumulant convention, cf. Remark 3.6). Odd-block classes vanish by the local law. Hence

$$m_5 = 1 + \underbrace{\frac{10}{3} + 10t_{\text{adj}} + 5t_{\text{opp}} + 5\Phi_4}_{=67/12 \text{ exactly}} + C_5, \quad m_6 = \frac{39}{4} + 6C_5 + \{2, 2, 2\} + \{4, 2\} + \{6\},$$

the rational parts being anchor-free (the t -geometry cancels: $10t_{\text{adj}} + 5\Phi_4 + 5t_{\text{opp}} = \frac{5}{4}$ and $30t_{\text{adj}} + 15t_{\text{opp}} + 15\Phi_4 = \frac{15}{4}$ identically — verified symbolically in the certification audit and kernel-compiled in the Lean package, §11(f)), with $\{2, 2, 2\} = \frac{131}{420}$ now *certified exactly* (Proposition 8.2), $\{4, 2\} = -0.0552(8)$ (the four-cycle connected object consumed jointly with a spectator pair), and $\{6\} = -0.0078(8)$ (grouped midpoint-grid ladders, §8.6). The former write-out items W1–W3 are written out in §8.4.

8.3 The pairing layer certified exactly

Proposition 8.2 (Exact pairing constants). *The four-cycle anchors and the three-chord classes are the exact rationals*

$$t_{\text{adj}} = \frac{7}{60}, \quad t_{\text{opp}} = \frac{1}{30}, \quad T_0 = \frac{3}{70}, \quad T_1 = \frac{1}{90}, \quad T_2 = \frac{1}{180}, \quad T_3 = \frac{1}{70},$$

whence $\{2, 2, 2\} = 5T_0 + 6T_1 + 3T_2 + T_3 = \frac{131}{420}$ (Touchard–Riordan multiplicities $5/6/3/1$).

Proof. Each constant is the integral over a box of $(1 - \text{spread})_+ \cdot \prod \min(|x_i|, 1)$ for an explicit walk (the chord-pattern walks of the Bell ledgers): a piecewise-polynomial integrand all of whose kink hypersurfaces are $\{0, \pm 1\}$ -combinations of the variables at levels $\{0, \pm 1\}$. Since 0 is a walk position and every variable is a position or a difference of two positions, the support lies in $[-1, 1]^d$ and the min-caps never bind. The integral is then computed *exactly* in rational arithmetic by iterated integration with full breakpoint enumeration: inner breakpoints have unit leading coefficients, so all collision and boundary resultants remain small-integer forms, and on every kink-free piece the integrand is a polynomial of degree at most $2/4/8$ (in the inner/middle/outer variable), integrated exactly by closed Newton–Cotes rules of order $5/7/9$ at rational nodes. Every piece is integrated twice (whole piece, and the two halves): for polynomials of degree below the rule order the two exact rationals must coincide, and any missed kink forces a discrepancy; no discrepancy occurred. t_{opp} and T_0 additionally admit short closed-form evaluations by orthant decomposition (Appendix C), agreeing with the algorithmic values. Code and outputs: `certification/exact-t222.py`. $\square$

Remark 8.3 (The anchor correction). The archived anchor $t_{\text{adj}} = 0.11717$ (rounds 69–71 grid) is corrected by Proposition 8.2 to $\frac{7}{60} = 0.11\bar{6}$: a +0.43% rectangle-rule bias at kinks. Nothing consumed downstream changes: the m_5 and m_6 rational layers are anchor-free (above), and the reconciliation of Remark 3.6 becomes exact. The formerly conjectural rational identifications of the three-chord classes are now theorems. Register items D7, D8.

8.4 The write-outs W1–W3

The three items carried as named write-outs in the audit ledger are discharged here; their pre-registered faces are in `certification/writeouts_verification.py` (all green; one face fired during construction and was converted — Appendix B).

Lemma 8.4 (W2: free-slot factorization). *In any b -cycle cell ledger, a slot whose lock is unconstrained factorizes exactly: the cell aggregate equals the contracted $(b-1)$ -cycle aggregate times the window mass $\sum_{m \in W} \Lambda(m)$ of the free slot, and $\sum_{m \in W} \Lambda(m) = |W|(1 + O(e^{-c\sqrt{\log X}}))$.*

Proof. The lock sum over the free slot’s offset is unrestricted, so the double sum factorizes by Fubini; on the walk-geometry side this is the frozen-slot identity (a singleton block repeats a walk value and leaves the overlap range unchanged — machine-exact, `face_o5_gate/face_b6_frame`). The mass evaluation is the prime number theorem with the classical error term [6, §18]; the consumption normalization divides it out. The face verifies both the exact Fubini identity on the live family and the PNT mass at height (1.00041 at $X = 2 \times 10^5$). $\square$

Lemma 8.5 (W3: triple-pair factorization). *The $\{2, 2, 2\}$ class aggregate of the sixth-moment ledger equals the product of the three pair main terms against the joint overlap volume, up to a deviation aggregate $\ll_A K J' V^2 (\log X)^{-A}$ in the consumed normalization.*

Proof. The six positions form three disjoint affine pair-chains. The p -local density factorizes over the three chains by the CRT and Proposition 6.1 (disjoint slots; the merger corrections are the pair events already booked in the local law), so the singular-series model is the product of the three twin series. Archimedean coupling enters only through the joint window overlap (the walk geometry O_6), which is the certified exact layer of Proposition 8.2. The deviation from the product of pair mains is controlled by three applications of the $k = 2$ instance of the spectral transport: by Proposition 8.1 the resolved variance factorizes over the three chains, each factor a windowed single prime sum, and the three-leaf closure of §7 applies to each factor with the exponents of Proposition 5.3. The face tracks the product model at 1.9% (median over four shift patterns) at $X = 6 \times 10^4$. $\square$

Proposition 8.6 (W1: the resolved multi-dimensional comb). *For a b -cycle with $b-1$ lock coordinates, the resolved dilated density $\rho(\tau_1, \dots, \tau_{b-1})$ obeys the multi-dimensional comb identification: the model comb is supported on the product CRT progressions (one congruence per coordinate, as in Lemma 7.4), with local factors the κ_b -values of Proposition 6.1 and twin-series factors off the moduli; and the resolved spectral L^2 -distance from the comb obeys the bound of Theorem 7.5 with the same exponents.*

Proof. By Proposition 8.1 the resolved variance is carried by the product $\prod_i |S_i|^2$ on the joint analysis grid, and the multi- τ transform is complete in each coordinate, so the $(b-1)$ -dimensional Parseval identity holds coordinatewise (verified exactly in the face, 1.2×10^{-16}). The frequency classification of §7 applies per factor: on all-major bins each factor obeys Lemma 7.1; any

minor factor is closed by Lemma 7.3; the comb identification in each coordinate is Lemma 7.4, whose CRT progression now runs per coordinate (complete orthogonality in each, since each transform covers its full period). The argument of Theorem 7.5 then repeats verbatim with $b - 1$ classification indices; the truncation exponents strengthen with k (Proposition 5.3). The previously stated bracketing (aggregate faces above, resolved four-cycle case below) is subsumed: the aggregate instances are the τ -summed cases, and the resolved four-cycle case is Lemma 7.4 itself. The resolved-comb face (worked 5-cycle family) shows the CRT class contrast directly. $\square$

8.5 The connected values

$C_5 = \int O_5 C_5^{\text{Urs}} = 0.0278(1)$: grouped midpoint-grid ladder (protocol of §8.6) 0.02784/0.02781/0.02780 at $dv = 0.05/0.04/0.032$, monotone settling; the O -weighted integration box is exact. The six-cycle connected value $\{6\} = -0.0078(8)$: midpoint ladder with the same protocol. (An earlier endpoint-grid evaluation $C_5 = 0.0275(5)$, $\{6\} = -0.0100(12)$ is retired: endpoint grids sign-flip on coarse steps at $b \geq 6$ — register item, 32nd conversion — while midpoint grids settle monotonically; the correction of the four-cycle anchors to their exact rationals, §8.3, was forced by the same audit.) Fluctuations are controlled by the factor-wise variance lemmas (the transports of Theorem 7.5; the resolved multi-dimensional comb exhibits are bracketed by the aggregate faces and the resolved four-cycle case — now Proposition 8.6). End-to-end faces: the assembled $m_5 = 5.6111(1)$ and $m_6 = 10.1657(18)$ both lie inside the sine-model bands (3.2).

Grading note. After the certification of §8.3, the constants still carried numerically are exactly three: C_5 , $\{4, 2\}$, $\{6\}$ — evaluations of explicit piecewise-polynomial integrals in dimensions 4, 4, 5, at the stated two-sided bands, not certified enclosures. By the dimension theorem of §8.6 each of the three is a finite rational combination of rational polytope volumes, hence *rational*; the exact-integration algorithm of Proposition 8.2 applies to them unchanged, the computation being heavier by the extra iterated levels and the ordered-partition count of the connected integrands. Their closed-form evaluation is the designated next computation (item N1, §11). The headline of this paper consumes these constants *with* their bands through the exact dual certificate of §9, so the claim is exactly: *the stated bounds hold provided the three connected constants lie within their stated bands.*

8.6 The convolution calculus, rationality, and the midpoint protocol

The connected constants admit a second, structural representation that both fixes their arithmetic nature and dictates the numerical protocol. Write W for the unit-square window autocorrelation and B for the sine-kernel box; the cumulant integrands assemble from iterated convolutions of these two profiles. Three results (derivations and verification gates in the reproduction package, the `cyclic_cumulant` and `midpoint_ladder` engines):

(i) *The trace/convolution identity.* At $b = 2$ the connected integral evaluates in closed form, $\int O_2 C_2 = 1 - \|W \star \widehat{B}\|^2 = \frac{1}{3}$ exactly — the calibration gate (F-TRACE) for everything below; the $b = 3, 4$ instances reproduce the exact pairing values of Proposition 8.2 to 3×10^{-14} .

(ii) *The partition-cyclic cumulant formula (F-CYC).* For the b -cycle connected class,

$$C_b(v) = \sum_{P \vdash [b]} (-1)^{|P|-1} \sum_{\text{cyclic orders of } P} \text{ov}(\text{merged-}w \text{ walk}),$$

the inner sum running over the distinct cyclic arrangements of the blocks and ov the overlap volume of the merged walk — validated against the direct Ursell integrands at $b = 3, 4, 5$ to 3×10^{-14} .

(iii) *Dimension and rationality.* In every (P, σ) -term of (ii) the walk constraints eliminate one integration variable $((m-1)$ -dimensionality), and the resulting domain is a finite union of polytopes cut by the unit-coefficient hyperplane family $\mathcal{H}$ of Appendix C(A1) with polynomial integrands of rational coefficients. Each term is therefore a rational number, and so are C_5 , $\{4, 2\}$ and $\{6\}$ — the same argument that produced $t_{\text{adj}} = \frac{7}{60}$ and $\{2, 2, 2\} = \frac{131}{420}$ in closed form. What separates the three remaining constants from Proposition 8.2 is only the piece count of the exact integration, not its kind (item N1, §11).

The midpoint protocol. The numerical ladders for (i)–(iii) revealed a grid pathology recorded as the registry’s 32nd conversion (Appendix B, D10): endpoint grids on the kinked integrands sign-flip at $b \geq 6$ on coarse steps, while midpoint grids — which never sample on the kink set $\mathcal{H}$ — settle monotonically. All connected constants of this paper are evaluated by grouped midpoint-grid ladders, calibrated at $b = 4$ against the exact values of Proposition 8.2 (agreement to the ladder’s own band), and the earlier endpoint-grid adoptions are retired.

9 Consumption and the proof of Theorem 1.1

With $(m_1, \dots, m_4) = (1, \frac{4}{3}, 2, \frac{13}{4})$ pinned two-sidedly (Theorem 3.4, Theorem 7.5, (2.5)), write $C_5 = 0.0278 + \delta$, $|\delta| \leq 0.0001$: the fifth- and sixth-moment constraints are

$$m_5 \geq M_5(\delta) := \frac{67}{12} + 0.0278 + \delta, \quad m_6 \leq M_6(\delta) := \frac{39}{4} + \frac{131}{420} + 6(0.0278 + \delta) - 0.0552 - 0.0078 + 0.0018,$$

the same δ entering both (correlated-band accounting: the $6C_5$ term of m_6 moves with the C_5 term of m_5 ; the independent m_6 band 0.0018 itemizes as $0.0002 + 0.0008 + 0.0008$ over the frozen-slot transport, $\{4, 2\}$ and $\{6\}$ — the former $\{2, 2, 2\}$ band contribution is retired by Proposition 8.2). The consumption is no longer a numerical linear program but an exact certificate:

Lemma 9.1 (Exact dual certificate). *Let $a = \frac{27}{50}$, $b = \frac{263}{200}$, $c = \frac{103}{50}$ and*

$$P(x) = \frac{[(x-a)(x-b)(x-c)]^2}{(abc)^2} = \sum_{k=0}^6 y_k x^k.$$

Then $P \geq 0$ on $\mathbb{R}$ (a perfect square), $P(0) = y_0 = 1$, $y_5 < 0$, $y_6 > 0$. Consequently, for any probability measure ν on $[0, \infty)$ — no support bound — with $m_1 = 1$, $m_2 = \frac{4}{3}$, $m_3 = 2$, $m_4 = \frac{13}{4}$, $m_5 \geq M_5$, $m_6 \leq M_6$:

$$\nu(\{0\}) \leq \int P d\nu = \sum_{k \leq 4} y_k m_k + y_5 m_5 + y_6 m_6 \leq \sum_{k \leq 4} y_k m_k + y_5 M_5 + y_6 M_6 =: w_0(M_5, M_6),$$

and w_0 is affine in (M_5, M_6) , hence maximized over the δ -band at a corner ($dw_0/d\delta = y_5 + 6y_6 < 0$, so the worst corner is $\delta = -0.0001$: $M_5 = \frac{168331}{30000}$, $M_6 = \frac{42701}{4200}$). There, exact rational arithmetic gives

$$w_0 = \frac{1153107070889}{11233957316589} = 0.10264\dots, \quad 1 - 2w_0 \geq 0.7947, \quad 1 - w_0 \geq 0.8973.$$

Proof. Nonnegativity and $P(0) = 1$ are immediate from the closed form; the sign conditions $y_5 = -2(a+b+c) \cdot (ab+bc+ca) \dots$ follow from the expansion with positive roots (displayed in the certification package; the denominator-cleared expansion, the corner evaluation, and the

final inequalities are kernel-checked in core Lean 4, `lean/RhGate/ Certificate.lean`, and verified independently in exact rational arithmetic by `certification/certify_lp.py`). The mass bound is the Chebyshev–Markov mechanism of Lemma 3.1 at degree six: $y_5 < 0$ converts the lower moment bound, $y_6 > 0$ the upper. The atoms (a, b, c) were extracted from the numerical LP extremal (a pinched four-atom measure) and rationalized; any rational triple gives a valid bound, so the LP solver leaves the trust base. $\square$

Theorem 1.1 follows through the consumption mechanism of §2. The rungs: m_4 alone gives 13/18, 31/36 (Corollary 7.7, certificate Q of Lemma 3.1, leading coefficient positive — also support-free). The Dirichlet L -function case is verbatim. All constants ineffective. $\square$

Corollary 9.2 (Sign-only rung). *If the three connected constants merely satisfy $C_5 \geq 0$, $\{4, 2\} \leq 0$ and $\{6\} \leq 0$ (with $C_5 \leq 0.1$), then, by the re-optimized exact certificate of Appendix C(A5),*

$$\liminf \frac{N_0^s}{N} \geq 0.7403, \quad \liminf \frac{N_d}{N} \geq 0.8701.$$

Sign certification is a far coarser target than an enclosure for the designated 4D/5D exact integration; the full payoff curve is tabulated in Appendix C(A5).

Remark 9.3 (The repaired intermediate rung; register D6). An earlier stage of this research quoted an intermediate rung (0.7295/0.8647) from the fourth- and fifth-moment chain alone, “consuming no sixth-moment information”. The certification audit found this rung unsound as stated: with only a *lower* bound on m_5 and no m_6 (or support) information, the origin-mass problem on unbounded support is not improved at all — mass $\varepsilon = c/\Lambda^4$ at height Λ boosts m_5 arbitrarily at vanishing cost to $m_1, \dots, m_4$, and the audit’s LP at support ranges 8/20/60/200 degraded $0.73144 \rightarrow 0.72580 \rightarrow 0.72340 \rightarrow 0.72257 \rightarrow \frac{13}{18}$. Any dual certificate for a lower fifth-moment bound needs a negative x^5 -coefficient dominated by a positive x^6 -coefficient, i.e. sixth-moment (or support) information is *necessary*. The rung is therefore demoted: it holds in the stated form only with the m_6 cap, in which case it is subsumed by Lemma 9.1. The full certificate is unaffected ($y_6 > 0$).

Merge-audit note on the band model. The audit first ran an *anti*-correlated variant (the δ of m_5 entering m_6 with opposite sign) and obtained 0.78972; the correlated form above is the correct accounting, since C_5 is a single unknown shifting both constraints together. Both runs are recorded in the register (D3) so the sensitivity of the headline to the band model is on the record: under the (incorrect, doubly conservative) anti-correlated accounting the headline would read 0.7897/0.8949.

10 Numerical verification

The standalone pipeline (`pipeline/`; `numpy`, `scipy`) reruns every gate in a few minutes. Central height table (sixteen coprime modulus families):

T	2400	4800	9600	19200	38400	57600	76800	153600
dispersion (10^{-3})	8.39	5.19	4.16	4.06	3.87	1.84	1.29	1.04
tracking (%)	1.5	1.0	0.7	0.9	0.8	0.4	0.2	0.2

(full-range geometric mean 0.706 per doubling: the pure averaging law $2^{-1/2}$). Further faces: local closure exact (zero violations, seven configurations); Parseval exact; per-bin Siegel–Walfisz face falling tenfold over the range; Bell(5)/Bell(6) gates exact; five- and six-fold product faces green at all heights; well-spacing zero violations; hybrid size control $24\text{--}31 \times$ height-stable; consumption LP grid-stable. The registry: 194 pre-registered checks, 160 passed, 34 fired and converted, all forward-recorded (the 31st–34th conversions belong to the corrected-constants pass and the Lean compiles, §11(f), Appendix B).

The certification audit of August 16, 2026 re-ran: the full moment-chain gate suite at $T = 9600$ (ALL PASS, 72s: dispersion 4.163×10^{-3} , tracking median 0.73%, SW RMS 1.599×10^{-3} , tightness $24.8\text{--}31.3 \times$, five-fold 0.06–1.29%, six-fold 0.01–0.73%, LP 0.79454/0.89727 at two grids under the since-retired endpoint constants; the corrected-constants rerun gives 0.79472/0.89736); the fourth-moment identity suite at $T = 9600$ (ALL PASS, 48s); the constant suite of §4 (`m1.suite.py validate`: 6/6 local factors, 100/100 vector sieve, anchors and cross-convention difference reproduced; core -0.00545 against archive -0.00544 ; α -truncation -0.00491); the sawtooth certification (`sawtooth.cert`: grid 1.8675, samples 1.8419/1.8494/1.8512, certificate 2.100); the continuum quadrature gates (`quadrature.cert`: geometric tail ratio 0.500 certified, $c_0 = -1.4228$ decade-stable); and the certificate suite of §3 (`certificate_verification.py` exact; `moment_lp_reopt` 0.0721; `mu3_ledger_gates`; `cumulant_engine` gates C_2 exact, $\int O_3 C_3 = 0$). An independent re-implementation of the consumption LP (different code, HiGHS backend) reproduced every rung as recorded in §9.

A second certification pass (same date) added: the exact pairing-layer integrals (`exact_t222.py`: all values of Proposition 8.2, dual-scheme identical rationals, the closed-form cross-checks, and the match of the conjectured rationals); the exact dual certificate (`certify_lp.py`: expansion, both band corners, final inequalities, all exact); the write-out faces (`writeouts_verification.py`: free-slot Fubini exact, window mass 1.00041, triple-pair product model median 0.9814, resolved Parseval 1.2×10^{-16} , resolved CRT contrast — one face fired during construction and was converted, the registry’s 30th); and the support-range probe behind Remark 9.3.

The corrected-constants pass re-ran: the grouped midpoint-grid ladders for C_5 , $\{4, 2\}$, $\{6\}$ (§8.6; checkpointed engines, $b = 4$ calibration exact against Proposition 8.2); the exact dual certificate at the corrected corners (`certify_lp.py`: expansion, both band corners, final inequalities, all exact, worst corner $w_0 = \frac{1153107070889}{11233957316589}$); the convolution trace gates (F-TRACE: $b = 2$ exact $\frac{1}{3}$, $b = 3, 4$ to 3×10^{-14}); and — for the first time in the program — a genuine Lean kernel compile of the identity layer on the maintainers’ toolchain (§11(f)).

Remark 10.1 (Effective faces, ineffective theorem). The faces are finite-height measurements with effective constants; the theorem’s constants are ineffective (Siegel). The faces verify the exact-identity layer, the model identifications, and the qualitative averaging behaviour the proof predicts; they cannot and do not verify the asymptotics. The two kinds of evidence are kept distinct throughout.

11 Verification status

Claim grade: certified-candidate. The chain = classical inputs (Siegel–Walfisz [6, 9], Vaughan, singular-series truncation, [11] for §4) + archived certified lemmas (bridge, glue, aggregation, D1 truncation, consumer interface, anchor certificates: §4) + the machine-verified identities of §§3, 6–8 + the numerically evaluated connected constants of §8.5 + the consumption layer of §2. Grades by layer:

(a) *Machine-exact or exact-rational, re-verified in the certification audit.* The certificate algebra (Lemma 3.1) and $\kappa(c)$ laws; the local closure (Proposition 6.1); the Parseval and product identities; the Bell ledgers and frozen-slot identities; the anchor-free 67/12 and 39/4; the 197/60 ledger arithmetic and the now fully exact two-bookkeeping reconciliation (Remark 3.6); the exact pairing-layer constants $t_{\text{adj}} = \frac{7}{60}$, $t_{\text{opp}} = \frac{1}{30}$, $\{2, 2, 2\} = \frac{131}{420}$ (Proposition 8.2, Appendix C); the exact dual certificate (Lemma 9.1) — the consumption step involves no LP solver, no grid, and no support bound; the sawtooth constant $C_{\text{saw}} = 2.10$ (grid-plus-analytic certification).

(b) *Certified-candidate analytic layer (pre-Lean, pre-review).* The nine-class μ_3 ledger (two nonstandard steps written out in the archived memo); the truncation discharge; the spectral proof of Lemma D (Theorem 7.5) — whose minor/mixed-bin size bookkeeping is classical in kind but carried at write-out grade with face-checked budgets — and its factor-wise transports to $k = 5, 6$, including the written-out W1–W3 (§8.4), which are proved at this layer’s grade; the covered-zone/band/glue chain of §4. Five adversarial audit rounds precede the frozen chain (archive rounds r93, r95, r100, r102).

(c) *Write-out items.* Discharged: W1 (Proposition 8.6), W2 (Lemma 8.4), W3 (Lemma 8.5), each with a pre-registered face. W2 is complete (elementary); W1 and W3 are proved at the grade of layer (b), since they consume the same spectral transports.

(d) *Numerical constants, not yet certified enclosures.* N1 (reduced, and now known rational): C_5 , $\{4, 2\}$, $\{6\}$ (grouped midpoint-grid ladders with stated bands, §8.5; each proved a rational number by the dimension theorem of §8.6, so certification means identifying a fraction, not enclosing a real); the designated route is the exact-integration algorithm of Proposition 8.2 (same kink structure; heavier by dimension and by the ordered-partition count — cluster scale, mechanical in kind). The *certificate family* of Appendix C(A5) makes the payoff graded and plug-in: any future enclosure yields an exact certificate, and a sign-only certification already yields 0.740/0.870. N2 (partially discharged): the quadrature’s slope is proved exactly 1, its constant is identified in closed form, $c_0 = \gamma - 2$ (Proposition C.1), and the geometric tail was already certified (r60); what remains computed is the mean oscillation $\overline{\text{osc}}$ inside c^* and the universality-collapse rate write-out. Only the conservative charge ε_{CL} is consumed anywhere (Lemma 4.10). Full certification of N1 upgrades Theorem 1.1 to the grade of Corollary 7.7; of N2, Theorem 4.1 to 0.7031/0.8515.

(e) *Imports.* [4]: the framework of §2 (matrix, block structure, tail, counting, first two moments) — a publicly posted preprint of this program, not independently peer-reviewed; its own analytic imports are published literature. Directly consumed published literature: [17] and [9, Thm. 5.12] (explicit formula), [16, Thm. 9.2] (zero density), [12, 1] (prime-side pair-correlation evaluation, unconditional in the band), [11], [8], [6].

(f) *Formalization status.* The identity layer has now been *compiled and kernel-checked* on the maintainers’ machine: core Lean 4 (toolchain `leanprover/lean4:v4.33.0`, no `mathlib`), package `lean/RhGate/`, build log included. Compiled content: the anchor-free assembly identities $m_4 = \frac{13}{4}$, $m_5\text{-part} = \frac{67}{12}$, $m_6\text{ pair-layer} = \frac{39}{4}$ as universally quantified rational identities (any anchors t_a, t_o — `Anchor.lean`); the consumption arithmetic $\frac{13}{18}, \frac{31}{36}$; and the coincidence-counting local law at $b = 4$, $p = 5$ by full kernel enumeration of all 125 lock classes (`LocalClosure.lean` — this theorem is *axiom-free*; the others use only `propext`, `Classical.choice`, `Quot.sound`). Zero `sorry`s. The compile itself fired a registered check and became the 33rd conversion: the seed files’ proofs invoked `ring` and `decide` on $\mathbb{Q}$, neither available in core Lean (`ring` is a `mathlib` tactic; core `Rat` arithmetic is not kernel-reducible, so `decide` stalls on `instDecidableEqRat`); all five affected proofs were repaired with the core `grind` tactic, statements unchanged (Ap-

pendix B). A second same-machine compile (August 16, 20:09) covered the certificate layer: `Certificate.lean` (the denominator-cleared expansion identity of Lemma 9.1, nonnegativity, normalization, both corner evaluations of w_0 , the corner-selection inequality $y_5 + 6y_6 < 0$ — so the *choice* of binding corner is itself kernel-checked — the headline inequalities $1 - 2w_0 \geq \frac{7947}{10000}$, $1 - w_0 \geq \frac{8973}{10000}$, the monic expansion of Appendix C(A6), the exact $\{2, 2, 2\}$ arithmetic and the instantiated anchor identities) and `LocalClosure2.lean` (the local law at $b = 4$, $p = 7$ and $b = 5$, $p = 5$: 343 and 625 lock classes by kernel `decide` on $\mathbb{N}$ — both *axiom-free*). This compile fired the registry a second time (the 34th conversion): `grind`’s ring/linear core treats a square $(\cdot)^2$ as an opaque atom and cannot close perfect-square nonnegativity; the repair is the *core* ordered-ring lemma `Lean.Grind.OrderedRing.sq_nonneg` (term-mode, statement unchanged, still *mathlib-free*). The pre-compile seed `SOSCertificate.lean` — whose headline constants carried the retired endpoint-grid corner — is removed from the package to preclude coexisting 0.7946/0.7947 headline theorems; its still-valid content is absorbed into `Certificate.lean`, and the draft is preserved in the program archive. Every statement is proxy-verified in exact rational arithmetic (`certify_lp.py`) or full Python enumeration (zero deviations). What Lean certifies is the rational-arithmetic layer alone: the moment pinning, the measure-theoretic Chebyshev–Markov step and the analytic chain remain graded as in layers (b)–(e). The truncation, spectral and consumption *analysis* remains unformalized; formalization of the analytic layer — which relies on the framework of [4] — is graded pending. The formalization package accompanying [4] is separate from this repository; the compiled coverage of this paper is exactly as stated above.

(g) *The gate*. Per the program’s verification gate, the record claim follows (i) the certified fractions N1 (and optionally N2), (ii) compiled Lean formalization of the identity and certificate layers with the queue’s Q1–Q6 (both are now compiled, §11(f); the analytic layer remains), and (iii) external review. No Riemann Hypothesis, zero-density input, pair-correlation conjecture, or Kloosterman technology is used anywhere. Nothing here bears on RH itself.

Acknowledgements

The mathematical development of this paper — the framework of §2, the certificate calculus, the spectral proof of Lemma D, the moment identifications, the convolution calculus, and the reproduction and Lean packages — was carried out by Claude (Anthropic), a large language model, working within a research program directed by the authors under the falsifier-first verification discipline described in §11 and Appendix B. The authors specified objectives, arbitrated decisions, compiled and audited the Lean package on their own toolchain, and take full academic responsibility for the content. This paper is built on the two-thirds theorem [4], a preprint of the same research program.

A The comb identification, with a worked pair

In the τ -frame the model of $\rho(\tau)$ is assembled from twin singular series: with $\mathfrak{S}$ the Hardy–Littlewood series $\mathfrak{S}(h) = 2C_2 \prod_{p|h, p>2} \frac{p-1}{p-2}$ (h even), window overlap counts V_m, V_n , and exact diagonal pieces,

$$\text{Model}(\tau) = \sum_{\substack{t \equiv 0 \pmod{b_2} \\ t \equiv \tau \pmod{b_1} \\ t \notin \{0, \tau\}}} \mathfrak{S}\left(\frac{t}{b_2}\right) \mathfrak{S}\left(\frac{t-\tau}{b_1}\right) V_m\left(\frac{t}{b_2}\right) V_n\left(\frac{t-\tau}{b_1}\right) + \text{diagonals}.$$

The CRT progression $t \equiv 0 \pmod{b_2}$, $t \equiv \tau \pmod{b_1}$ is the *complete* orthogonality of the joint frame: a shift t contributes iff it is simultaneously a b_2 -lattice twin shift on the m -side and, offset by τ , a b_1 -lattice twin shift on the n -side; no incomplete sums occur because the τ -transform runs over the full period. The local factors of $\mathfrak{S}$ at the primes $p \mid b_1 b_2$ are exactly the κ -values of Proposition 6.1 at $b = 4$; for $p \nmid b_1 b_2$ they are the twin factors.

Worked pair $(b_1, b_2) = (5, 3)$: $M = 15$, and for $\tau \equiv \tau_0$ the surviving shifts are the single residue class $t \equiv t_0(\tau) \pmod{15}$ with $t_0 = 3 \cdot (2\tau \bmod 5) \bmod 15$ (here $3^{-1} \equiv 2 \pmod{5}$). For example $\tau = 1$ gives $t \equiv 6 \pmod{15}$: the model sums $\mathfrak{S}(t/3)\mathfrak{S}((t-1)/5)V_m V_n$ over $t = 6, 21, 36, \dots$ and the negative branch, plus the diagonal $t = \tau$ term when $3 \mid \tau$ and $t = 0$ when $5 \mid \tau$ (neither here). This is precisely what `face_dispersion` constructs; its agreement with the measured profile at 0.2–4% across all sixteen families and eight heights is the numerical face of Lemma 7.4. At cycle lengths 5 and 6 the same complete-orthogonality logic gives the product-model combs of §8 (aggregate faces green at 0.1–2.5%).

B Provenance, research record, and discrepancy register

The research record

The mathematics of this paper was developed in an interactive research program (August 2026) whose working discipline is falsifier-first: every structural claim was registered with a numerical check before adoption, every firing was converted on the forward record, and the complete trail — round logs, memos, claim ledgers, correction history, scripts — is preserved in the program archive that accompanies the reproduction package. The four stages of the chain (§3, §4, §§5–7, §§8–9) were each closed under adversarial audit rounds before the certification audit of §11 froze the chain presented here.

Conversions that shaped this text

Six of the registry’s fired-and-converted checks shaped the mathematics directly: the evaluation-transfer repair (envelopes are never transported to limits; r61); the model-subtraction correction of the reduced covariance (r80); the hybrid size budget (two firings, r88); the τ -frame lesson (fixed-cutoff arc statistics never self-average; the Parseval identity entered here; r90); the incomplete-sum slip of the progression frame (r95, Remark 7.6); and the removal of an unproven sixth-moment cap from an internal ladder rung (the intermediate values quoted in Theorem 1.1 consume exactly what their chains prove). The 29th conversion (r103) is the Lean-draft inverse omission of §6.

Discrepancy register of the certification audit (August 16, 2026)

- D1. Earlier draft of §4, assembly paragraph: printed $\Lambda_2(0) = 0.156293$; the closed form at $\bar{R} = 0.0066$ gives 0.154193, and the theorem constants 0.6916/0.8458 are consistent with the latter. Corrected in §4; theorem unaffected. *Impact on the main theorem: none.*
- D2. An earlier draft attributed the bands 5.573 ± 0.113 , 10.02 ± 0.27 to measurement “on 2×10^7 zeta zeros”; the reproduction package derives exactly these numbers from GUE double-window Richardson extrapolation (`gue_bands.py`, `m5m6_bands.py`), and no zeta-zero dataset of that size exists in the record. This paper attributes the bands to the

- sine-model/GUE measurement (3.2). If a zeta-zero measurement exists it is not in the record and is not consumed. *Impact: attribution only; no constant changes.*
- D3. Consumption LP band model: correlated vs anti-correlated δ -accounting (0.79454 vs 0.78972); the correlated form is correct and is what `face_lp_full` implements; both on record (§9). *Impact: the anti-correlated variant is doubly conservative; headline unchanged.*
- D4. The m_4+m_5 rung: an earlier stage quoted 0.7295/0.8647; the audit’s independent LP gives 0.73144/0.86572 under the same constraints at bounded support. Superseded by D6: the rung is demoted regardless. *Impact: none on the main chain (the demoted rung is not consumed).*
- D5. Formalization language: earlier program documents described [4] as “Lean-formalized”; that statement concerns the package accompanying [4], of which no compiled artifact exists in this repository (§11(f)). The compiled coverage of this paper is exactly as graded in (f). *Impact: grading language only.*
- D6. The intermediate m_4+m_5 rung (0.7295/0.8647, an earlier stage) is unsound on unbounded support without sixth-moment information: the audit’s LP degrades to 13/18 as the support range grows (0.73144 $\rightarrow$ 0.72257 at $R = 8 \rightarrow 200$), and a dual certificate for a fifth-moment lower bound necessarily consumes x^6 -information. Rung demoted and repaired (Remark 9.3); the main chain is unaffected ($y_6 > 0$ in Lemma 9.1). *Impact: removes an intermediate rung; headline unaffected.*
- D7. The archived four-cycle anchor $t_{\text{adj}} = 0.11717$ (rounds 69–71; quoted in earlier drafts and `m1.suite`) carries a +0.43% rectangle-rule grid bias: the exact value is $\frac{7}{60} = 0.11667$ (Proposition 8.2). Downstream totals are unaffected (the m_5/m_6 rational layers are anchor-free); the saturated-split triple 0.2677/0.0156/−0.0177 becomes exactly $\frac{4}{15}/\frac{1}{60}/-\frac{1}{60}$ (Remark 3.6), in better agreement with the cumulant engine’s independent −0.01663. *Impact: none on the rational layers (anchor-free); corrects the archived anchor.*
- D8. The rational identifications $T_0, \dots, T_3 = \frac{3}{70}, \frac{1}{90}, \frac{1}{180}, \frac{1}{70}$, recorded in the audit ledger as a conjecture and not consumed, are now proven (Proposition 8.2); $\{2, 2, 2\} = \frac{131}{420}$ is consumed exactly and its band contribution retired. *Impact: tightens m_6 ; contributes to the headline.*
- D9. The continuum-quadrature constant $c_0 = -1.4228$ (r60, “computed, not fitted”) is identified in closed form: $c_0 = \gamma - 2$ (Proposition C.1). The audit located a $d = 2$ discrepancy between the archive’s tensor-sieve γ -array and the raw Euler closed form ($\tilde{\gamma}_2 = C_2 - 1$ vs $\gamma_2 = C_2$, shifting the constant by exactly -2); this is the archive’s parity-mean convention (the $q = 2$ arc block booked in $\bar{D}$), not an error — the raw-convention constant is γ , and both conventions are now documented with the conversion. The slope-1 law of the partial sums, previously a measurement (0.99996), is proved exactly. *Impact: $N2$ partial discharge; main theorem unaffected.*
- D10. The endpoint-grid ladders behind the previous connected constants fired their refinement face: at $b \geq 6$ and coarse steps the endpoint evaluation *sign-flips* ($b = 6$ at $dv = 0.4$: +0.036 against the true −0.010), and the archived adoptions $C_5 = 0.0275(5)$, $\{4, 2\} = -0.0548(10)$, $\{6\} = -0.0100(12)$ inherited a systematic shift. Grouped *midpoint*-grid ladders (§8.6), calibrated exactly against Proposition 8.2 at $b = 4$, settle monotonically and give the corrected $C_5 = 0.0278(1)$, $\{4, 2\} = -0.0552(8)$, $\{6\} = -0.0078(8)$ consumed

in §9 (registry’s 32nd conversion; the 31st is D7’s anchor correction, independently forced by the same audit). *Impact: direct — these are the constants the headline consumes; the theorem value moved $0.7946 \rightarrow 0.7947$ under the correction.*

- D11. The Lean draft files claimed core-decidability of their $\mathbb{Q}$ -statements (“**decide** on $\mathbb{Q}$ ”) and used **ring**; the maintainers’ compile showed both unavailable in core Lean 4.33 (**ring** is **mathlib**; **instDecidableEqRat** does not kernel-reduce). Five proofs repaired with **grind**, statements unchanged; build then succeeded (registry’s 33rd conversion; §11(f)). *Impact: proofs only; statements unchanged.*
- D12. The second compile fired on **cert_nonneg**: **grind** assigns the square $(\cdot)^2$ as an opaque atom (its diagnostics guess the atom’s value) and cannot derive perfect-square nonnegativity; repaired with core Lean **Grind.OrderedRing.sq_nonneg** (registry’s 34th conversion). The audit also found the superseded seed **SOSCertificate.lean** silently outside the build roots while declaring identically named headline theorems at the retired constants; the file is removed from the package (content absorbed; the draft is preserved in the program archive) and the build now covers every **.lean** file in the directory. *Impact: proofs and packaging only; statements unchanged.*
- D13. The citation relation to [4] was reversed twice on the forward record: an intermediate revision absorbed the framework into the text with proofs (when [4] was not yet publicly posted), and the present revision restores [4] as the cited foundation and removes the duplicated proofs, keeping statements only (§2). *Impact: attribution and exposition; no mathematical content changed.*

The registry’s 30th conversion occurred during the certification audit: the first version of the W1 resolved-comb face used raw multiples of $b_1 b_2$ as the on-progression and fired; the corrected face implements the per-coordinate CRT classes (§8.4). Forward-recorded in **writeouts_verification.py**. The 31st–34th are D7, D10, D11 and D12 above.

Reproduction

A consolidated reproduction checklist accompanies this paper (**REPRODUCTION.md**): environment, one command per suite, expected outputs with the August 16, 2026 rerun values, and the mapping from every quoted constant to the script that regenerates it. Total runtime of the full set is under fifteen minutes on one CPU (the session package’s optional long gates under thirty; the exact $\{2, 2, 2\}$ certification under two hours).

C The exact certification layer

A1. The exact-integration algorithm

Every pairing-layer constant is an integral $\int_{[-1,1]^d} (1 - \text{spread}(p_1, \dots, p_b))_+ \prod_i |x_i| dx$ where the walk positions p_j are $\{0, 1\}$ -combinations of the chord variables and 0 is a position (so the support forces $|x_i| \leq 1$ and the $\min(|\cdot|, 1)$ caps of the C_2 factors never bind). The integrand is piecewise polynomial; its kink set is contained in the hyperplane family

$$\mathcal{H} = \{p_i - p_j = c\}_{i < j, c \in \{0, \pm 1\}} \cup \{x_i = 0, \pm 1\},$$

all with coefficients in $\{-1, 0, 1\}$ and unit leading coefficients in the innermost variable. Iterated integration: the inner integral is computed exactly between the enumerated inner breakpoints; because leading coefficients are units, the breakpoints are themselves small-integer linear forms in the outer variables, and the piece structure of a partial integral changes only on (i) direct kinks, (ii) collisions of two inner breakpoints, (iii) breakpoint–boundary collisions — an explicitly enumerated resultant family at each level. On every kink-free piece the (partial) integrand is a polynomial of degree at most $2/4/8$ in the inner/middle/outer variable, integrated exactly by closed Newton–Cotes rules of order $5/7/9$ at rational nodes, in exact rational arithmetic throughout. *Self-check*: each piece is integrated by the same rule on the whole piece and on its two halves; for a polynomial of degree below the rule order both are the exact integral, so the two rationals must be *identical*, and a kink missed by the enumeration forces a discrepancy on the containing piece. No discrepancy occurred at any level of any run. Independent cross-checks: closed-form orthant evaluations (A2), and the float grid ladders of the archive (10^{-6} -agreement with the certified rationals).

A2. Closed-form cross-checks

T_0 (positions $\{0, v, w, u\}$): in the all-positive orthant $\int_{[0,1]^3} (1 - \max(v, w, u))v w u = \frac{1}{56}$ (order and integrate); a one-negative orthant gives $\int_0^1 t(1-t)^5/20 dt = \frac{1}{840}$; summing the orbit, $T_0 = 2 \cdot \frac{1}{56} + 6 \cdot \frac{1}{840} = \frac{3}{70}$. t_{opp} (positions $\{0, v, v+w, w\}$): the same decomposition gives $\frac{1}{30}$ exactly, confirming the archived exact anchor. t_{adj} : the two same-sign quadrants give $\frac{1}{20}$ each and the two mixed quadrants $\frac{1}{120}$ each: $t_{\text{adj}} = \frac{14}{120} = \frac{7}{60}$ (not the archived grid value 0.11717; register D7).

A3. The certified values

$$t_{\text{adj}} = \frac{7}{60}, \quad t_{\text{opp}} = \frac{1}{30}, \quad T_0 = \frac{3}{70}, \quad T_1 = \frac{1}{90}, \quad T_2 = \frac{1}{180}, \quad T_3 = \frac{1}{70}, \quad \{2, 2, 2\} = \frac{131}{420},$$

with $\Phi_4 = -\frac{1}{60}$, saturated $\{2, 2\} = \frac{4}{15}$, plateau $= \frac{1}{60}$, connected model $= -\frac{1}{60}$ (Remark 3.6). Code: `certification/exact_t222.py` (classes run separately; dual-scheme identical-rational check at every piece).

A4. The continuum-quadrature constant in closed form (N2, partial discharge)

Proposition C.1 ($c_0 = \gamma - 2$). *Let γ_d denote the free coefficients of the partial-sum identity (Lemma 4.7) at the $(1, 1)$ class. Then γ_d has the exact Euler closed form*

$$\gamma_{2m} = C_2 \prod_{p|m} \frac{1}{p(p-2)} \quad (m \text{ odd squarefree}), \quad \gamma_d = 0 \text{ otherwise,}$$

and, in the raw convention, $\sum_{d \leq M} d\gamma_d = \log M + \gamma + o(1)$; in the archive convention (the $q = 2$ arc block booked inside the parity mean, $\tilde{\gamma}_2 = \gamma_2 - \beta_2 = C_2 - 1$),

$$c_0 = \gamma - 2 = -1.4227843 \dots$$

Proof. (i) The closed form: from the arc identity (Proposition 4.2) at $b_1 = b_2 = 1$, $\beta_q = \mu(q)^2/\varphi(q)^2$, and $\gamma_d = \sum_e \mu(e)\beta_{de}$ factorizes over primes with local weights $f_0(p) = 1 - (p-1)^{-2}$, $f_1(p) = (p-1)^{-2}$, $f_{v \geq 2} = 0$; $f_0(2) = 0$ forces $2 \parallel d$ and the product assembles to the display (C_2 the twin-prime constant). (ii) The constant: $\sum_{d \leq M} d\gamma_d = 2C_2 \sum_{m \leq M/2} \prod_{p|m} \frac{1}{p-2}$, whose

Dirichlet series is $\zeta(s+1)E(s)$ with $E(0) = \prod_p(1 - \frac{1}{p}) \prod_{p>2}(1 + \frac{1}{p-2}) = \frac{1}{2C_2}$ (slope $2C_2E(0) = 1$ exactly) and $E'(0)/E(0) = \sum_p \frac{\log p}{p-1} - \sum_{p>2} \frac{\log p}{p-1} = \log 2$ (the p -sums cancel pairwise except $p = 2$); the double-pole residue gives $2C_2[E(0)(\log \frac{M}{2} + \gamma) + E'(0)] = \log M + \gamma$. (iii) The convention shift: the archive tensor sieve (`tail_bound`) and the closed form agree at *every* d except $d = 2$, where the archived class- d replacement books $\tilde{\gamma}_2 = C_2 - 1$ (the $q = 2$ block moved to the parity mean); this shifts $\sum d\gamma_d$ by exactly -2 . Numerical verification (`n2.c0.certification.py`): the raw sum converges to γ with deviations $9.4 \times 10^{-4}/6.3 \times 10^{-5}/9.2 \times 10^{-6}$ at $M = 10^4/10^5/2 \times 10^7$; the archived measurement -1.42279 at $M = 1.6 \times 10^7$ matches $\gamma - 2$ to 10^{-5} ; the partial-sum identity closes with its $d > M$ tail (LHS $-$ RHS $-$ tail $= -0.010$ at $M = 2 \times 10^5$, $D = 2 \times 10^7$, consistent with the remaining truncation). $\square$

Consequently the “computed constants” of Lemma 4.10 reduce to: slope $= 1$ (proved), $c_0 = \gamma - 2$ (closed form), the certified geometric tail T_Γ (archive `r60`, ratio 0.500), and the single remaining computed piece $\overline{\text{osc}}$ inside c^* . N2 is thereby partially discharged; what remains is the $\overline{\text{osc}}$ identification (or certified enclosure) and the universality-collapse rate write-out.

A5. The N1 certificate family and its payoff curve

The exact dual certificate re-optimizes against any hypothetical certified enclosures $C_5 \in [c_-, c_+]$, $\{4, 2\} \leq u_{42}$, $\{6\} \leq u_6$ (correlated accounting: the single unknown C_5 enters m_5 and m_6 together; the worst case is a corner of the affine family). Exact rational certificates (`certificate_family.py`, atoms re-optimized per scenario):

scenario (what the 4D/5D exact integration certifies)	$1 - 2w_0 \geq$	$1 - w_0 \geq$
current bands (± 0.0001 ; u_{42}, u_6 at band tops)	0.79472	0.89736
enclosure width 0.01	0.77418	0.88709
enclosure width 0.04	0.74299	0.87149
signs only: $C_5 \geq 0$, $\{4, 2\} \leq 0$, $\{6\} \leq 0$	0.74031	0.87016
no connected information	13/18	31/36

Sensitivity of the shipped certificate (exact): $\partial(\text{headline})/\partial c_- = +7.32$, $\partial/\partial c_+ = -5.61$, $\partial/\partial(u_{42}+u_6) = -0.93$. In particular a *sign-only* certification of the three connected constants — far coarser than an enclosure — already certifies 0.740/0.870, and the designated 4D/5D exact computation has a graded payoff at every achievable width. Computational spec from the certified 3D runs: the exact integrator’s cost is (pieces $\times$ nodes) per level (≈ 40 – 400 per level in 3D at ≤ 90 kink forms); the C_5 and $\{4, 2\}$ integrands carry ≤ 541 resp. $\leq 75 \times$ (spectator) overlap terms on the same $\{0, \pm 1\}$ -kink family, one extra level ($\{6\}$: two), placing the runs at cluster scale rather than laptop scale.

A6. The exact dual certificate

The polynomial of Lemma 9.1 expands (numerator, monic) as

$$x^6 - \frac{783}{100}x^5 + \frac{975601}{40000}x^4 - \frac{19203237}{500000}x^3 + \frac{1599367847}{50000000}x^2 - \frac{16571397771}{1250000000}x + \frac{534950348409}{250000000000},$$

with $(abc)^2 = \frac{534950348409}{250000000000}$, so $y_0 = 1$, $y_5 = -\frac{783}{100}(abc)^{-2} < 0$, $y_6 = (abc)^{-2} > 0$. Corner evaluations at the corrected constants (exact): $w_0(\delta = -0.0001) = \frac{1153107070889}{11233957316589}$ (binding), $w_0(\delta = +0.0001) = \frac{1151185570889}{11233957316589}$.

Construction of the atoms. The triple $(a, b, c) = (\frac{27}{50}, \frac{263}{200}, \frac{103}{50})$ is extracted from the primal extremal measure of the moment problem, then rationalized; its validity does not depend on the extraction. In detail: (i) the discretized linear program (maximize origin mass subject to $m_1, \dots, m_4$ pinned, $m_5 \geq M_5$, $m_6 \leq M_6$; grid $dx = 0.001$) has optimal measure supported on four atoms, $\{0, 0.5385, 1.3195, 2.0645\}$ at the binding corner — complementary slackness forces the optimal dual polynomial to vanish doubly at the three non-origin atoms, which dictates the shape $P = [(x-a)(x-b)(x-c)]^2/(abc)^2$; (ii) the atoms are rationalized to low-height fractions within 5×10^{-3} of the extremal support; (iii) *any* positive rational triple yields a valid certificate — P is a perfect square, so $P \geq 0$; $P(0) = 1$ by the normalization; the monic numerator has x^5 -coefficient $-2(a+b+c) < 0$ and x^6 -coefficient $1 > 0$, so $y_5 < 0 < y_6$ automatically — hence the LP solver, the grid, and the extraction all leave the trust base, and only the exact rational evaluation of w_0 at the chosen triple remains, which is Lemma 9.1 (kernel-checked). The rounding cost of the rationalization is 6×10^{-6} in the headline (LP optimum 0.794716 at the same corner against the certified 0.794710). Verification: `certification/certify_lp.py`; Lean: `lean/RhGate/Certificate.lean`, kernel-compiled (§11(f)): the denominator-cleared identity, this monic expansion, nonnegativity, both corners, the corner-selection sign $y_5 + 6y_6 < 0$ and the headlines; zero `sorry`s.

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